# UVI Ocean Glider Lab Piloting Training
# Day 1: The Basics

Authors: Joshua Soll and Travis Hamlin
Adapted from: Slocum Training, Joe Gradone (Rutgers U)

# Roadmap

Slocum Gliders

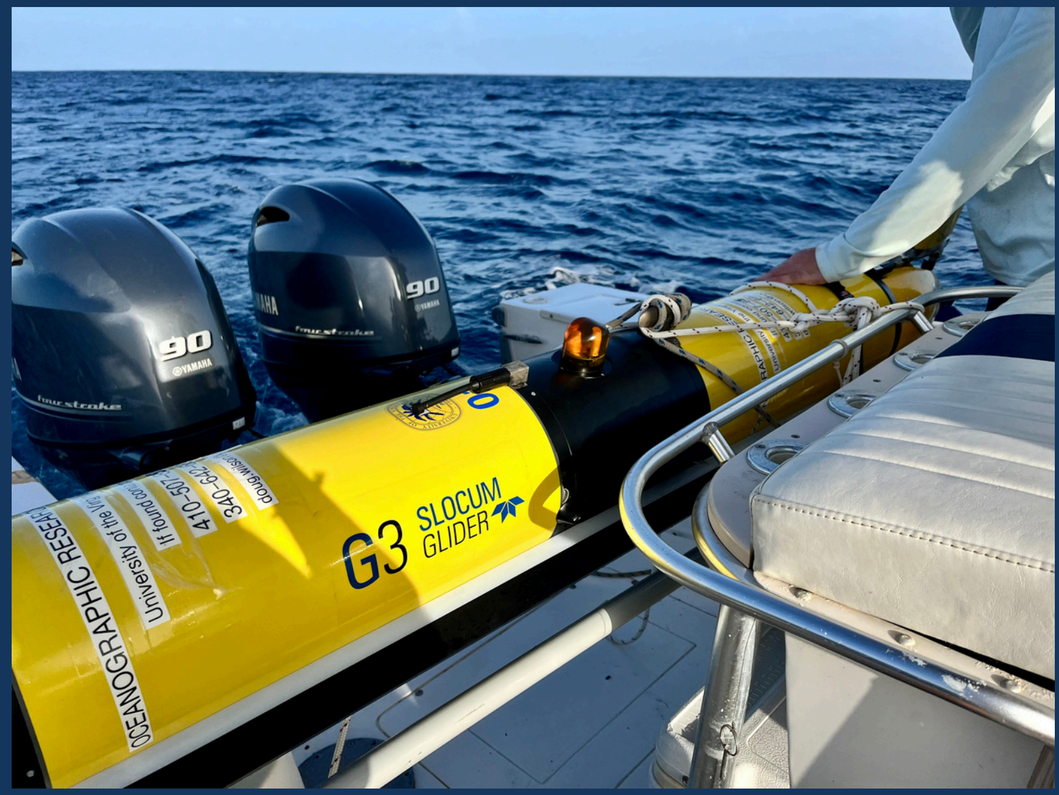
Piloting Considerations

Using SFMC (Mission Control)

Oddities, Warnings, Errors, and Aborts

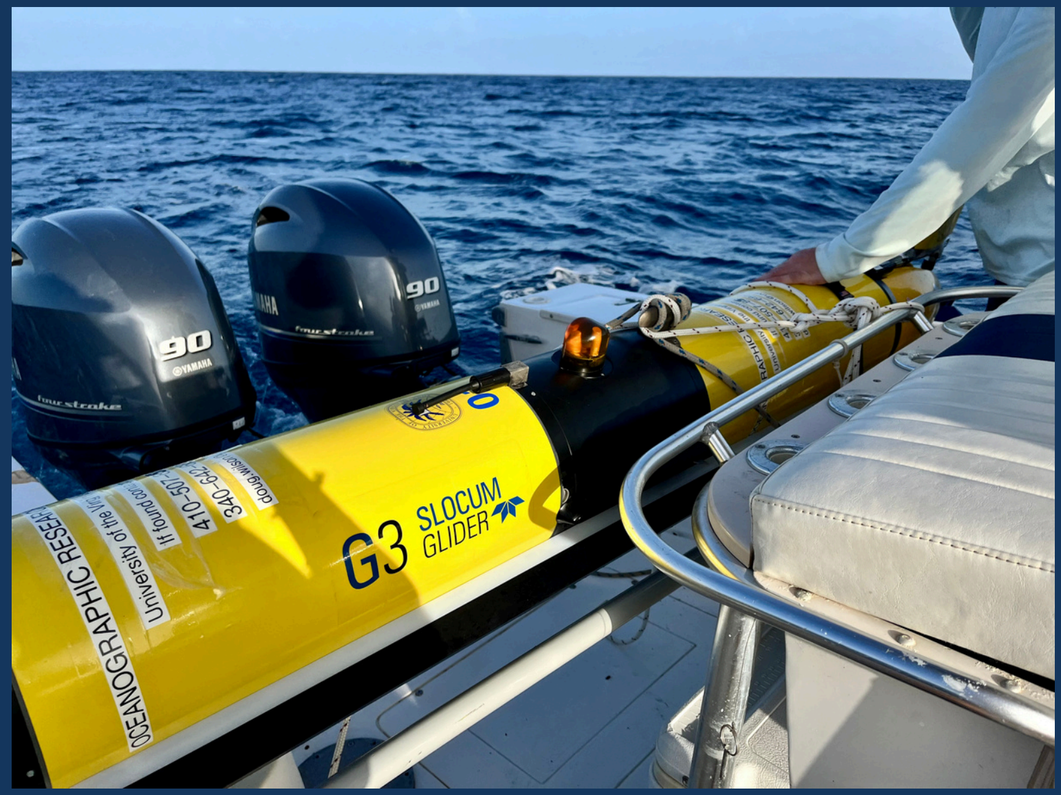
# What is an AUV/Glider?

- AUV = Autonomous Underwater Vehicle



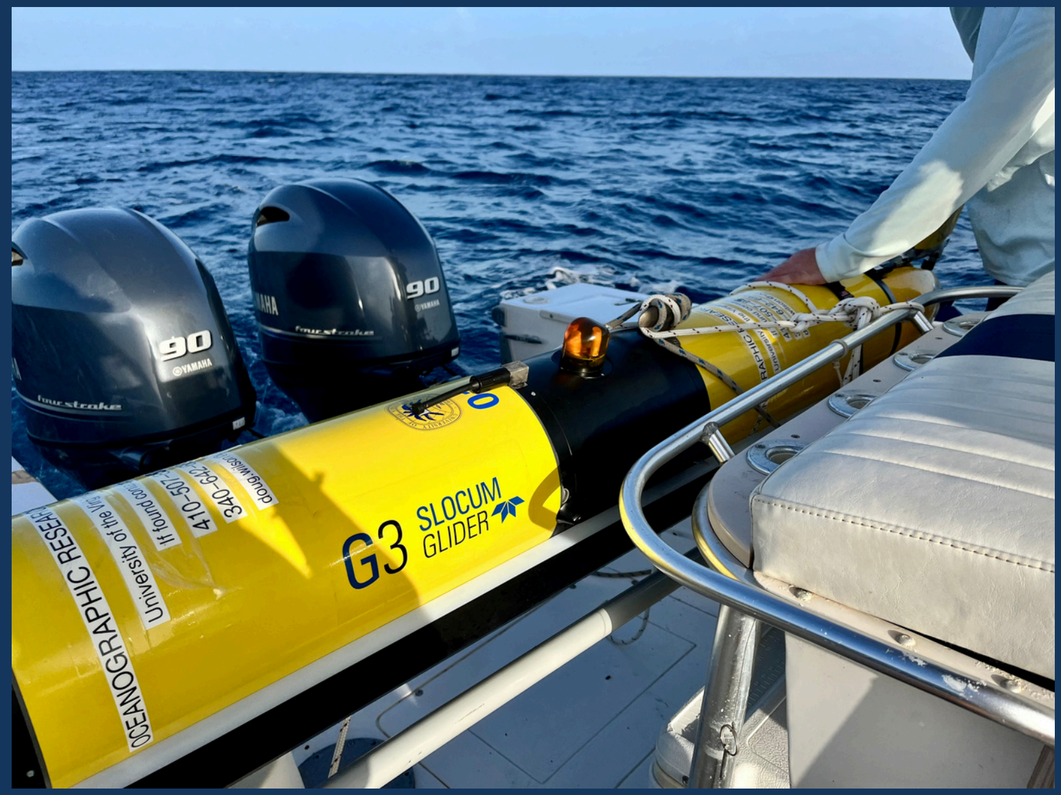
# What is an AUV/Glider?

- AUV = Autonomous Underwater Vehicle
- Remotely controlled through satellites (i.e., Iridium) and can transfer data



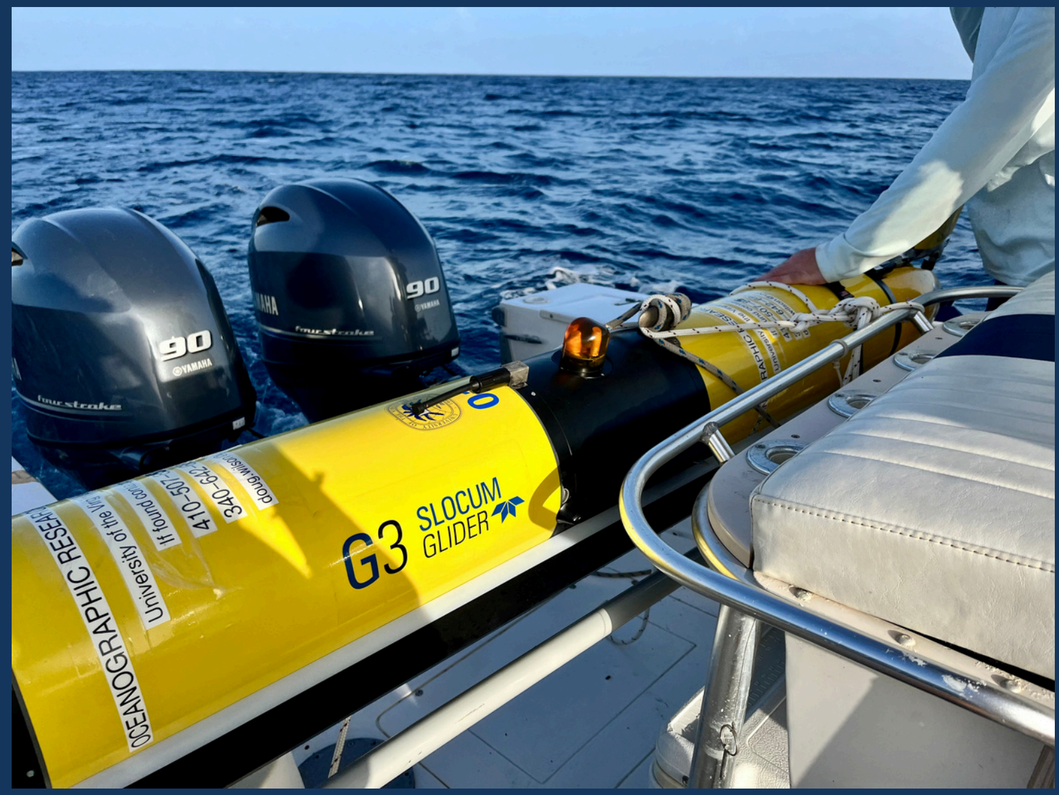
# What is an AUV/Glider?

- AUV = Autonomous Underwater Vehicle
- Remotely controlled through satellites (i.e., Iridium) and can transfer data
- Changes buoyancy to dive and climb, uses wings for lift to “glide” in water



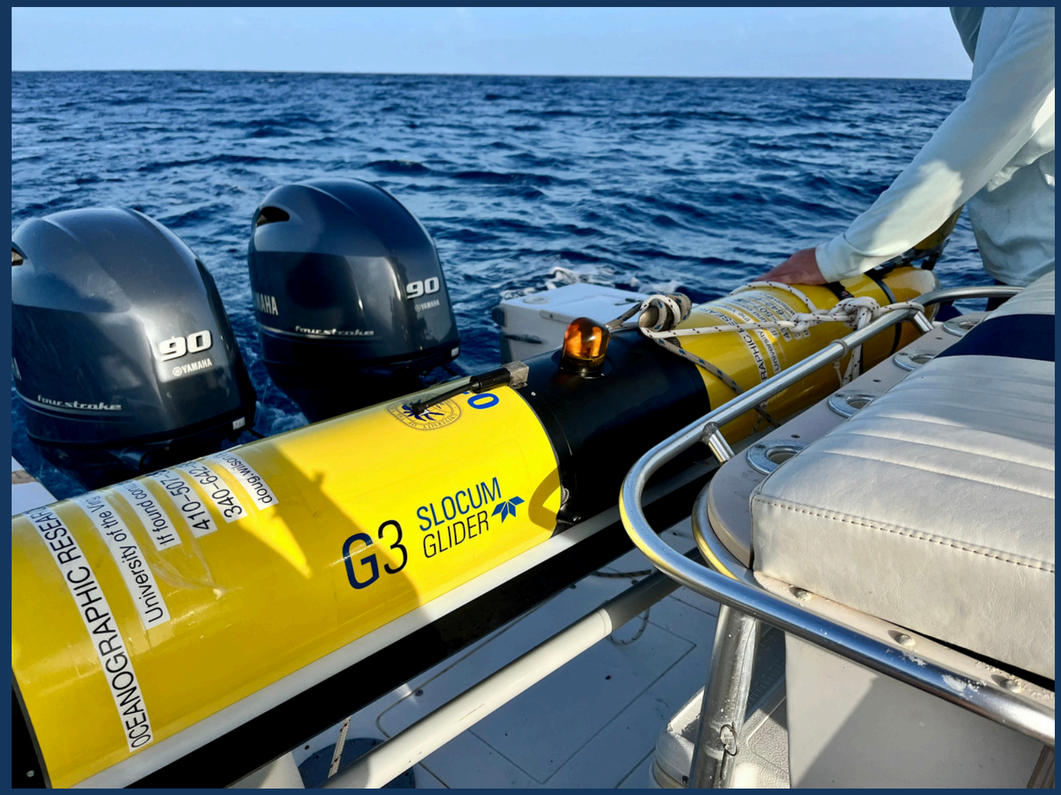
# What is an AUV/Glider?

- AUV = Autonomous Underwater Vehicle
- Remotely controlled through satellites (i.e., Iridium) and can transfer data
- Changes buoyancy to dive and climb, uses wings for lift to “glide” in water
- Can spend long periods of time in the field and underwater, traveling and collecting data

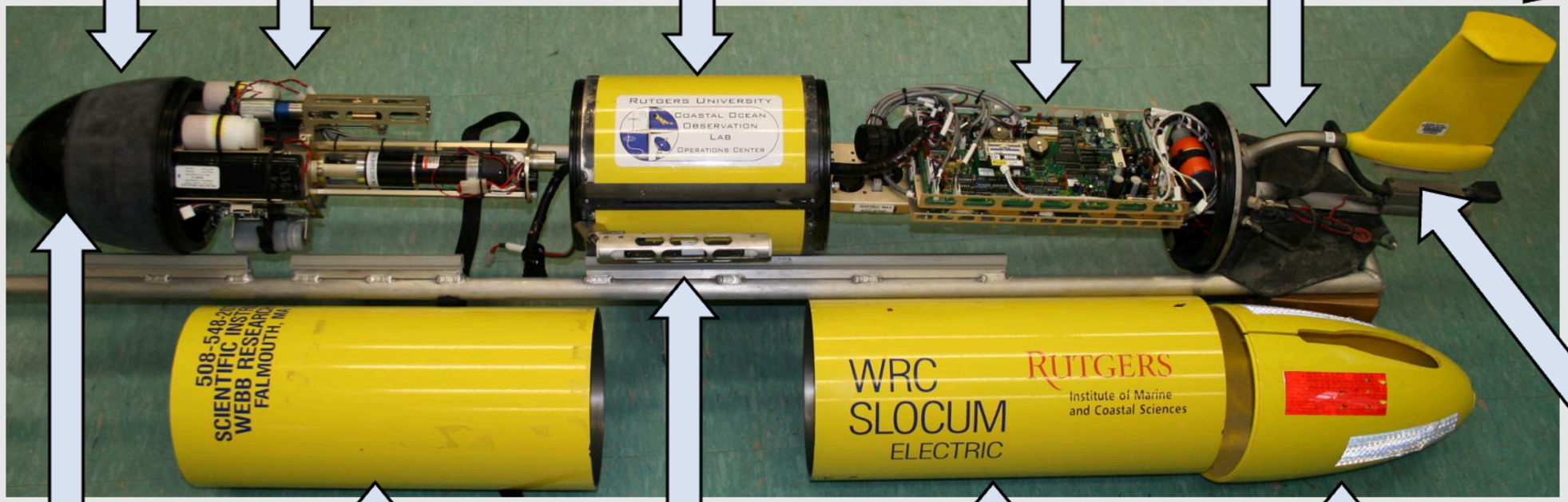


# What is an AUV/Glider?

- AUV = Autonomous Underwater Vehicle
- Remotely controlled through satellites (i.e., Iridium) and can transfer data
- Changes buoyancy to dive and climb, uses wings for lift to “glide” in water
- Can spend long periods of time in the field and underwater, traveling and collecting data
- Highly configurable – variety of sensors, battery types, compartments
    - CTD, Optics (Chlorophyll, Turbidity, CDOM), DO, Currents, Suspended Sediment, Waves, Active Acoustics, (Currents, Biomass), Passive Acoustics (Sound)



# Glider Anatomy




Buoyancy Pump

Pitch Motor

Science Payload Bay

Control Board (compass, processor, etc)

Air Bladder

Antenna (Iridium/GPS, Freewave, & Argos)

Radome/Digifin (Rudder, strobe, ejection weight, vacuum plug, power plug)

Altimeter

Fore Hull

CTD

Aft Hull

Cowling

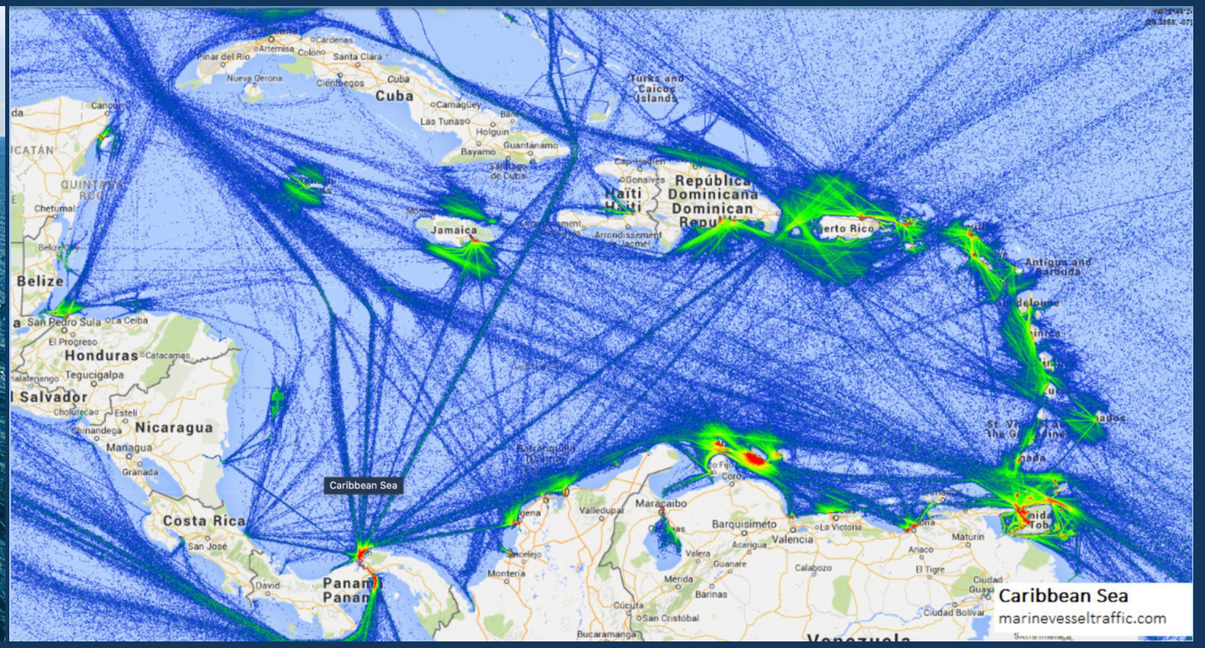
**Figure: Joe Gradone**

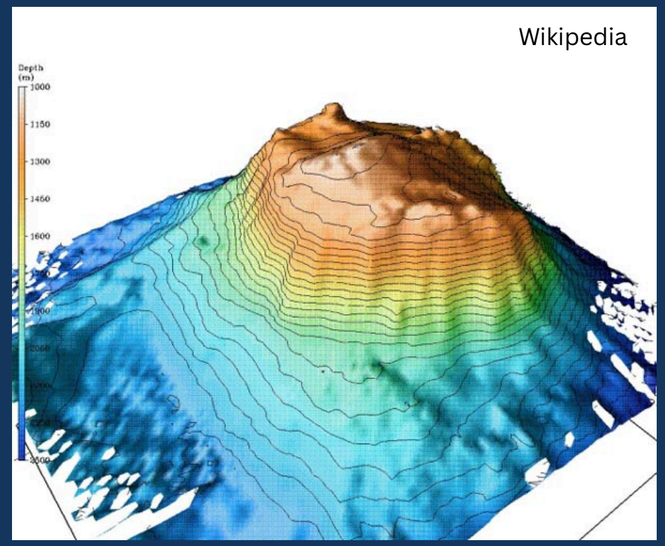
# Piloting Considerations

# Water Space Management

Always risks to glider when in the field. Be aware of:

- Tides
- Currents
- Shipping lanes
- Recreational traffic
- Fishing grounds
- Bathymetry (seamounts, slopes, etc)



# Glider Power Draw

- You cannot track actual battery life of glider batteries

# Glider Power Draw

- You cannot track actual battery life of glider batteries
- Coulomb counter is like “gas tank” of glider but don’t know exact capacity

# Glider Power Draw

- You cannot track actual battery life of glider batteries
- Coulomb counter is like “gas tank” of glider but don’t know exact capacity
- Manufacturer provides rough maximum amp-hrs for each battery type (alkaline, lithium primaries, lithium rechargeables)

# Glider Power Draw

- You cannot track actual battery life of glider batteries
- Coulomb counter is like “gas tank” of glider but don’t know exact capacity
- Manufacturer provides rough maximum amp-hrs for each battery type (alkaline, lithium primaries, lithium rechargeables)
- Track amp-hr usage over time, try avoid reaching actual estimated max amp-hr capacity. Average glider uses 5-9 amp-hrs/day.

# Glider Power Draw

- You cannot track actual battery life of glider batteries
- Coulomb counter is like “gas tank” of glider but don’t know exact capacity
- Manufacturer provides rough maximum amp-hrs for each battery type (alkaline, lithium primaries, lithium rechargeables)
- Track amp-hr usage over time, try avoid reaching actual estimated max amp-hr capacity. Average glider uses 5-9 amp-hrs/day.
- Can also watch battery voltage and use old voltage curves to estimate battery life

# Glider Power Draw

- You cannot track actual battery life of glider batteries
- Coulomb counter is like “gas tank” of glider but don’t know exact capacity
- Manufacturer provides rough maximum amp-hrs for each battery type (alkaline, lithium primaries, lithium rechargeables)
- Track amp-hr usage over time, try avoid reaching actual estimated max amp-hr capacity. Average glider uses 5-9 amp-hrs/day.
- Can also watch battery voltage and use old voltage curves to estimate battery life

**Alkaline Batteries**

- Least endurance
- Single use
- Most affordable
- Lightest
- ~168 total amp-hrs

**Lithium Rechargeables**

- Medium Endurance
- Requires battery management system (BMS) on board
- ~234-326 amp-hrs

**Lithium Primaries**

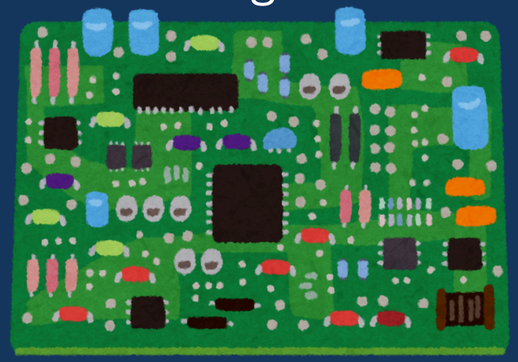
- Highest endurance
- Single use
- Most expensive
- ~600-870 amp-hrs

# Glider Operation

# File System

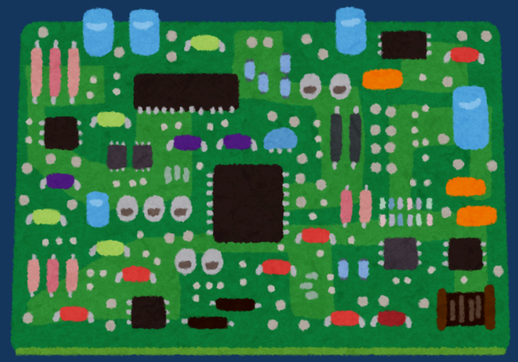
- Glider has two computers: Flight and Science computer

Flight



Controls glider operation, comms, diagnostics

Science

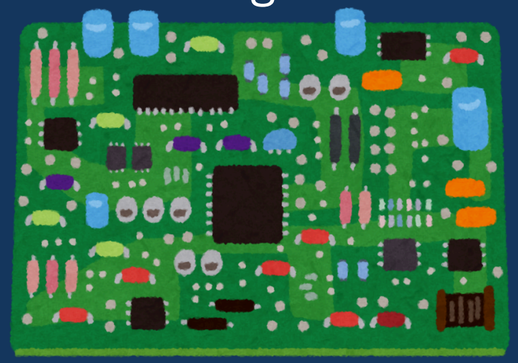


Controls science sensor data collection (CTD, O2, etc.)

# File System

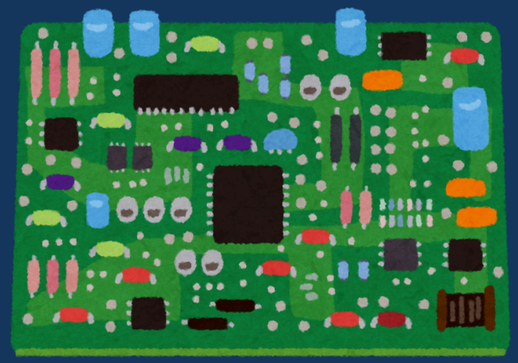
- Glider has two computers: Flight and Science computer
- Each holds a file directory with files that tell the glider what it is (i.e., what kinds of components it has) and what to do (i.e., where to go, how deep to dive, what data to collect)

Flight



Controls glider operation, comms, diagnostics

Science

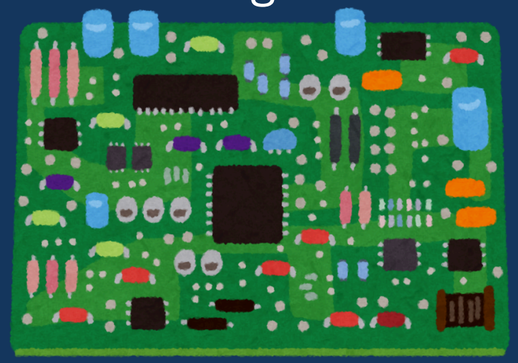


Controls science sensor data collection (CTD, O2, etc.)

# File System

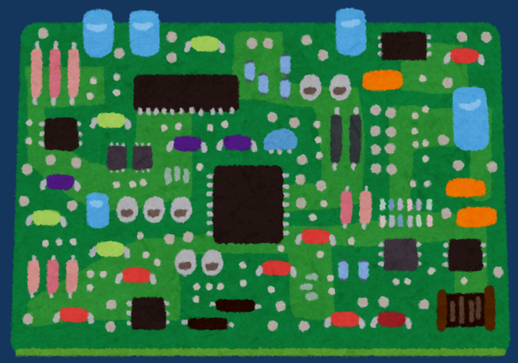
- Glider has two computers: Flight and Science computer
- Each holds a file directory with files that tell the glider what it is (i.e., what kinds of components it has) and what to do (i.e., where to go, how deep to dive, what data to collect)
- Highly configurable to mission goals.

Flight



Controls glider operation, comms, diagnostics

Science

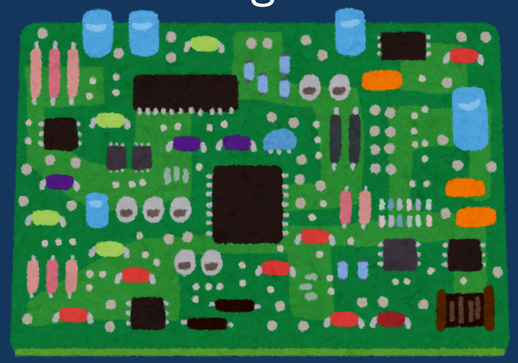


Controls science sensor data collection (CTD, O2, etc.)

# File System

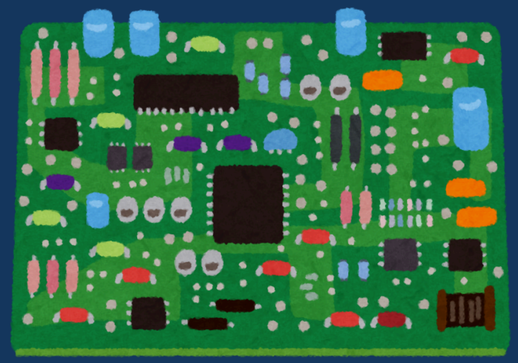
- Glider has two computers: Flight and Science computer
- Each holds a file directory with files that tell the glider what it is (i.e., what kinds of components it has) and what to do (i.e., where to go, how deep to dive, what data to collect)
- Highly configurable to mission goals.
- Some files should be left unchanged while some you change frequently

Flight



Controls glider operation, comms, diagnostics

Science

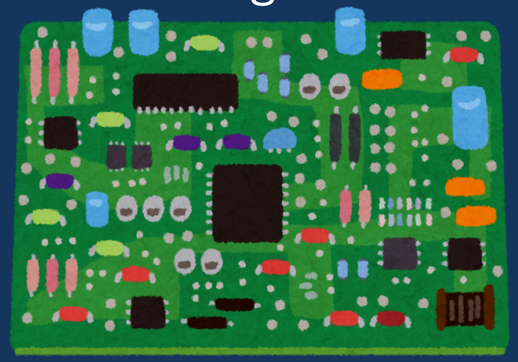


Controls science sensor data collection (CTD, O2, etc.)

# File System

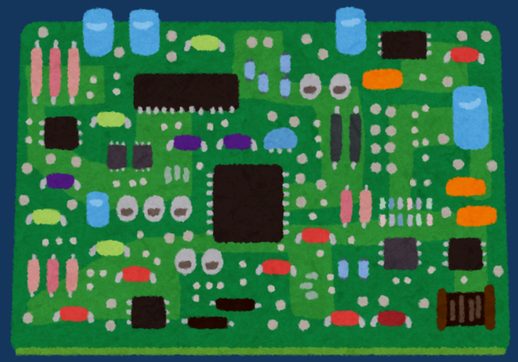
- Glider has two computers: Flight and Science computer
- Each holds a file directory with files that tell the glider what it is (i.e., what kinds of components it has) and what to do (i.e., where to go, how deep to dive, what data to collect)
- Highly configurable to mission goals.
- Some files should be left unchanged while some you change frequently
- Some files relate to other files, so if you change one, you may have to change others.

Flight



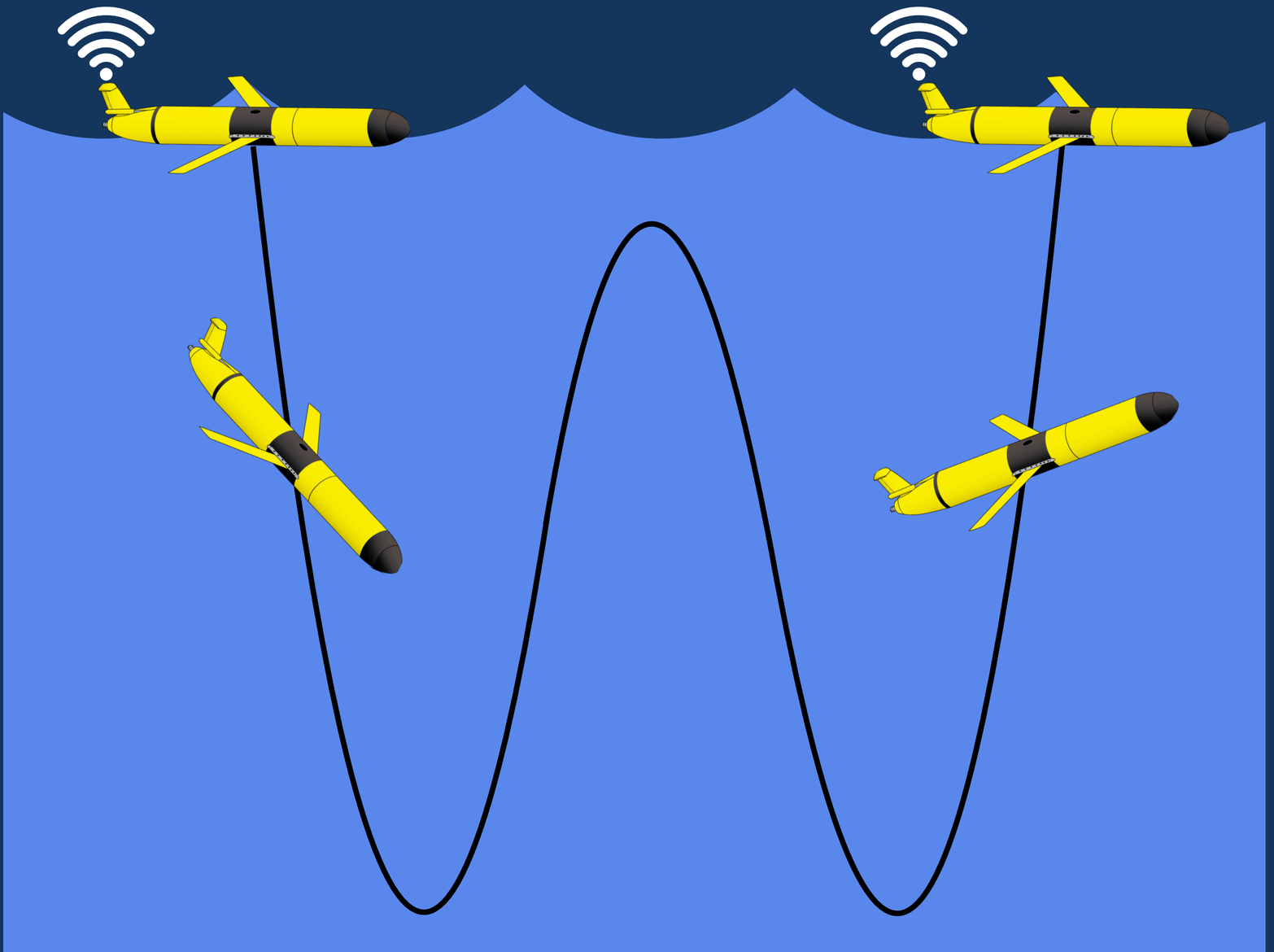
Controls glider operation, comms, diagnostics

Science

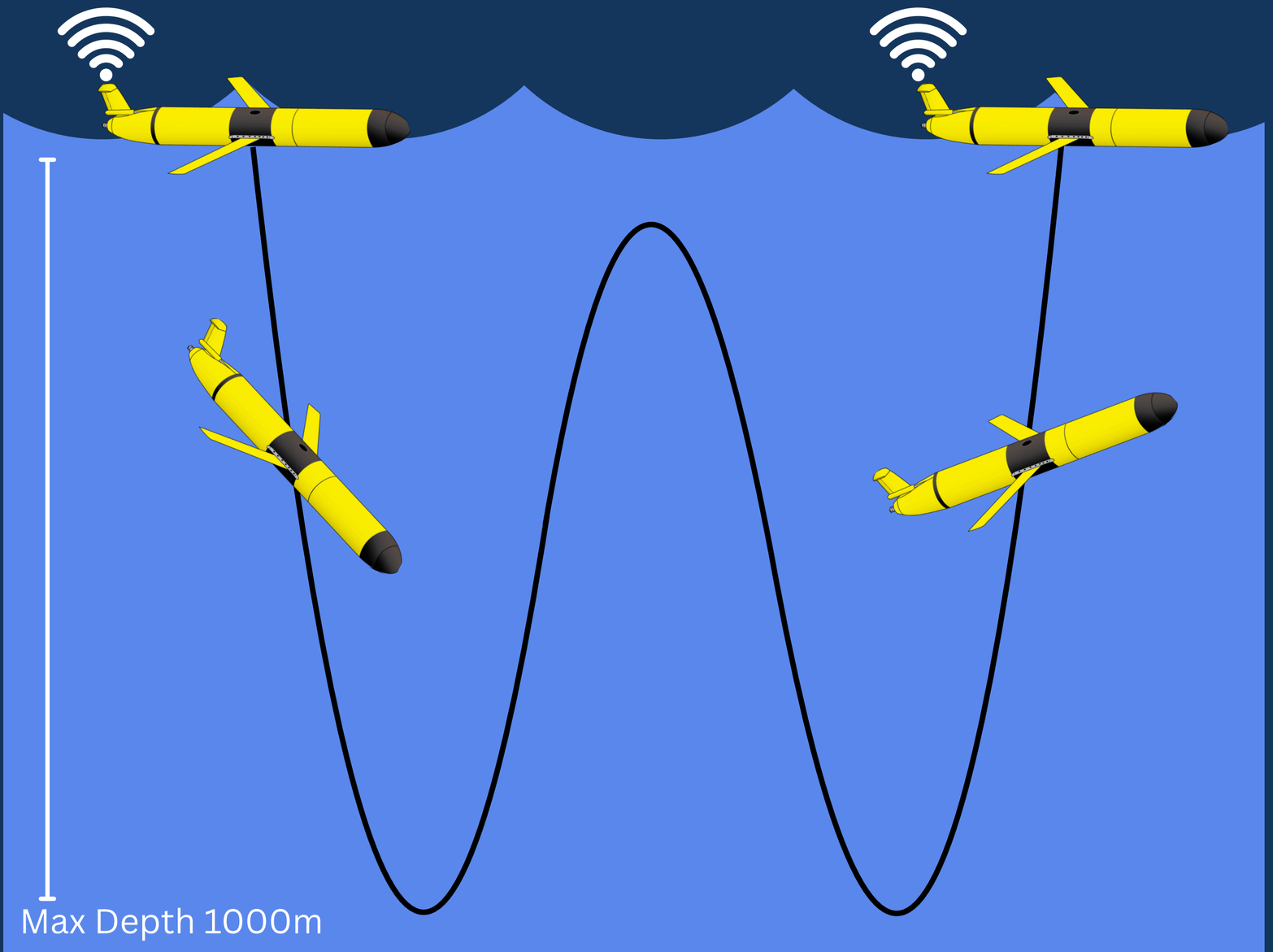


Controls science sensor data collection (CTD, O2, etc.)

# High-Level Glider Transit

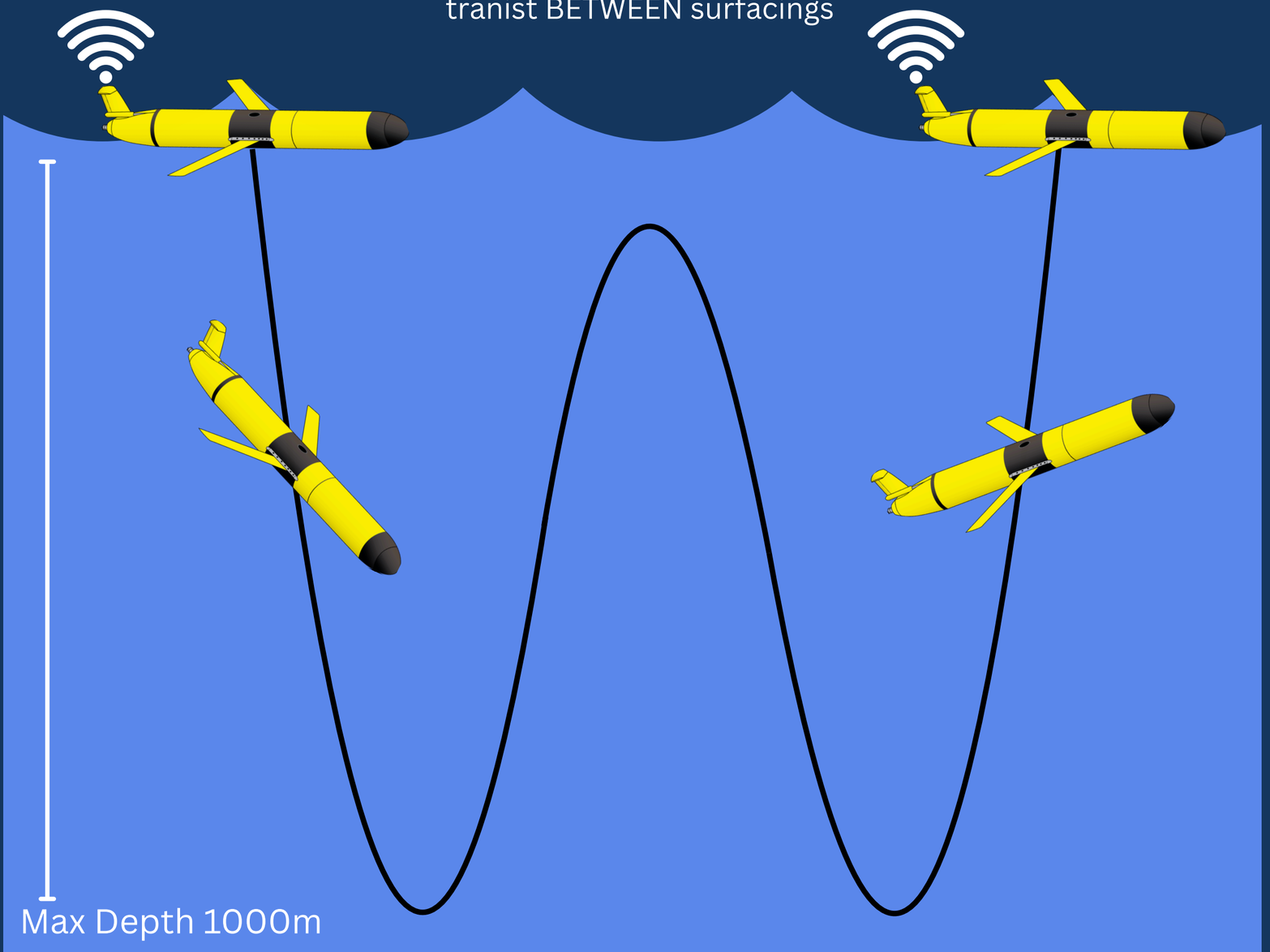


# High-Level Glider Transit



# High-Level Glider Transit

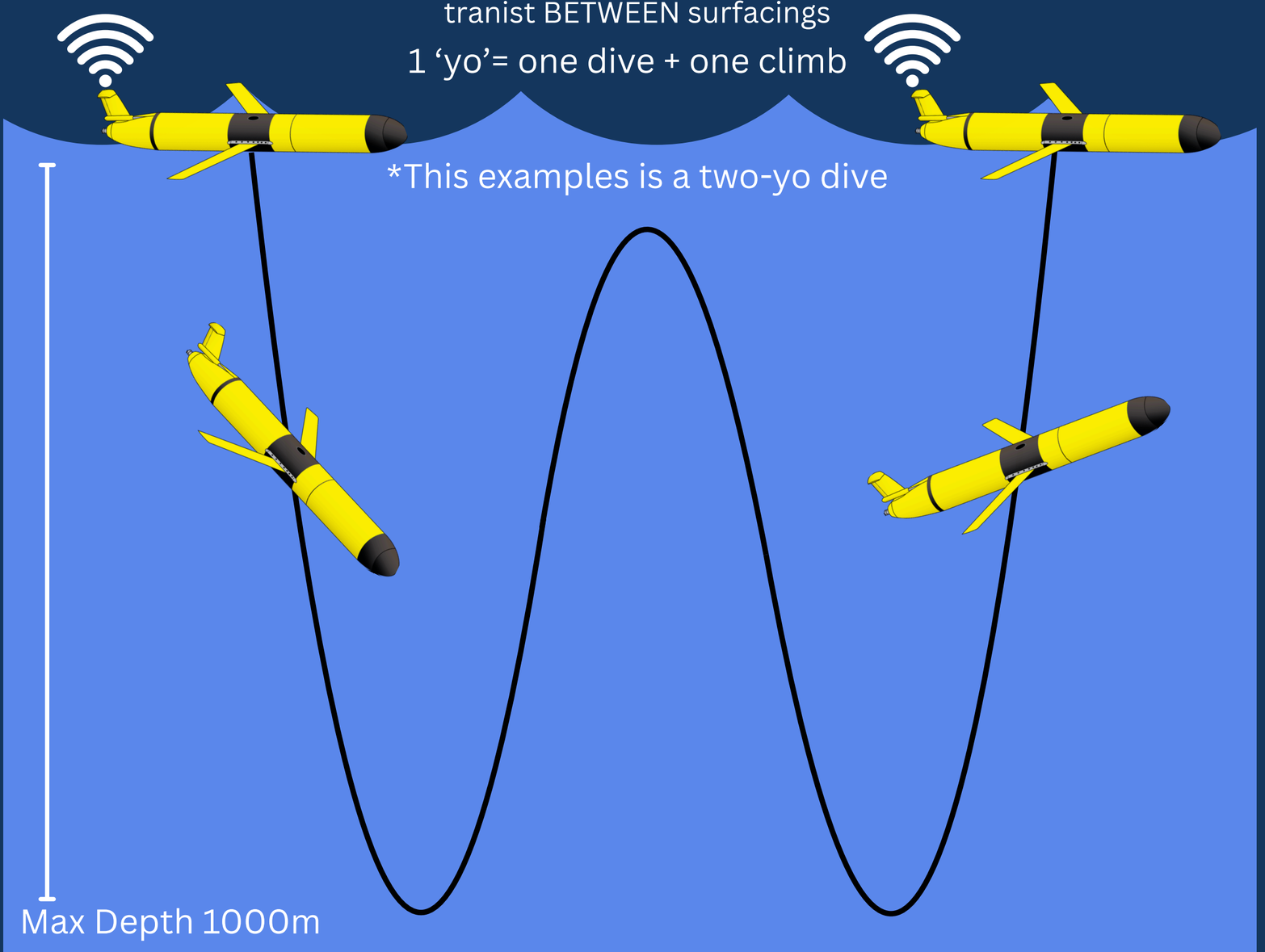
Segment = Portion of glider tranist BETWEEN surfacings



# High-Level Glider Transit

Segment = Portion of glider tranist BETWEEN surfacings

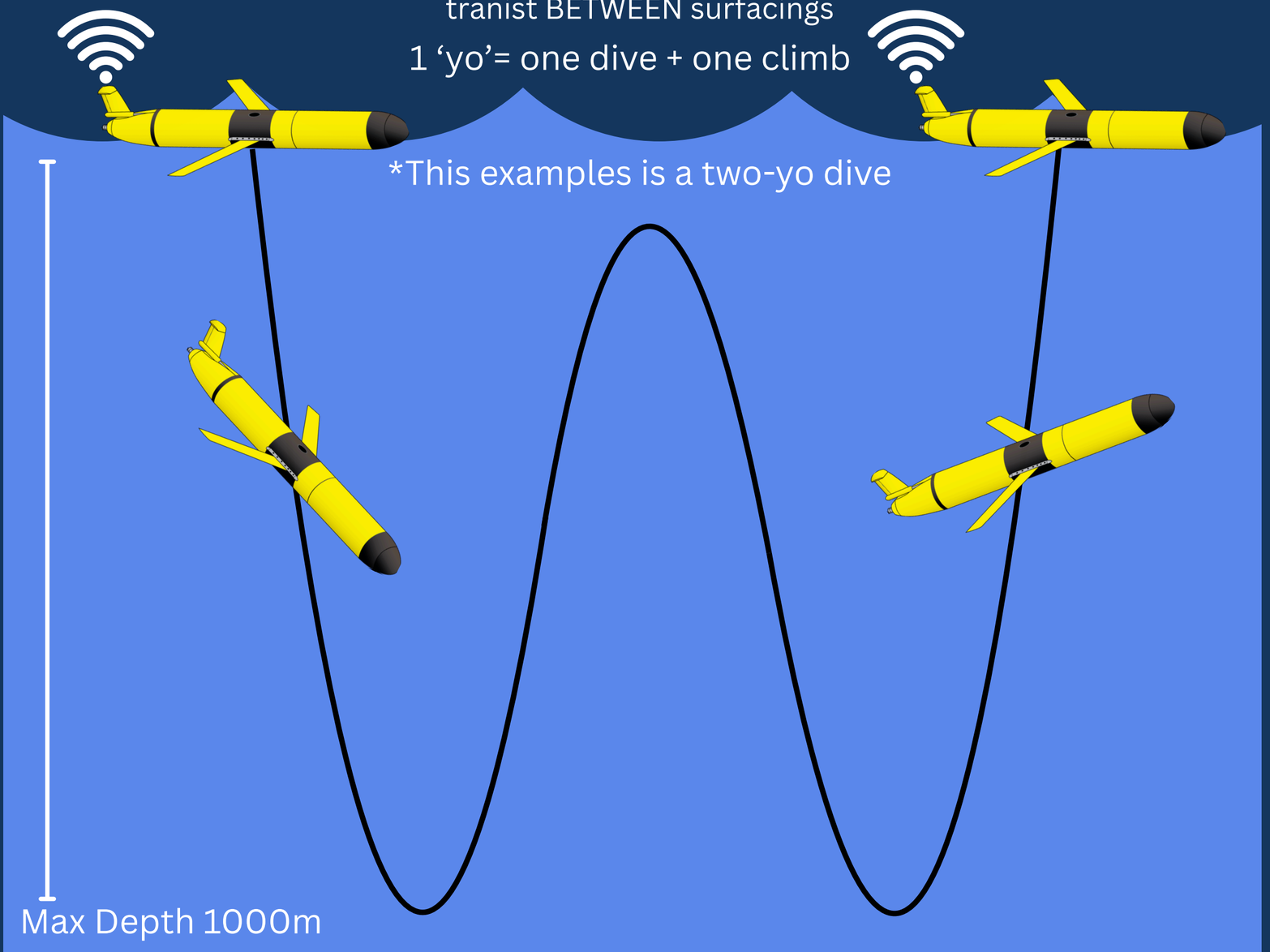
1 ‘yo’= one dive + one climb



# High-Level Glider Transit

Segment = Portion of glider tranist BETWEEN surfacings

1 ‘yo’= one dive + one climb




<u>Buoyancy driven, where:</u>

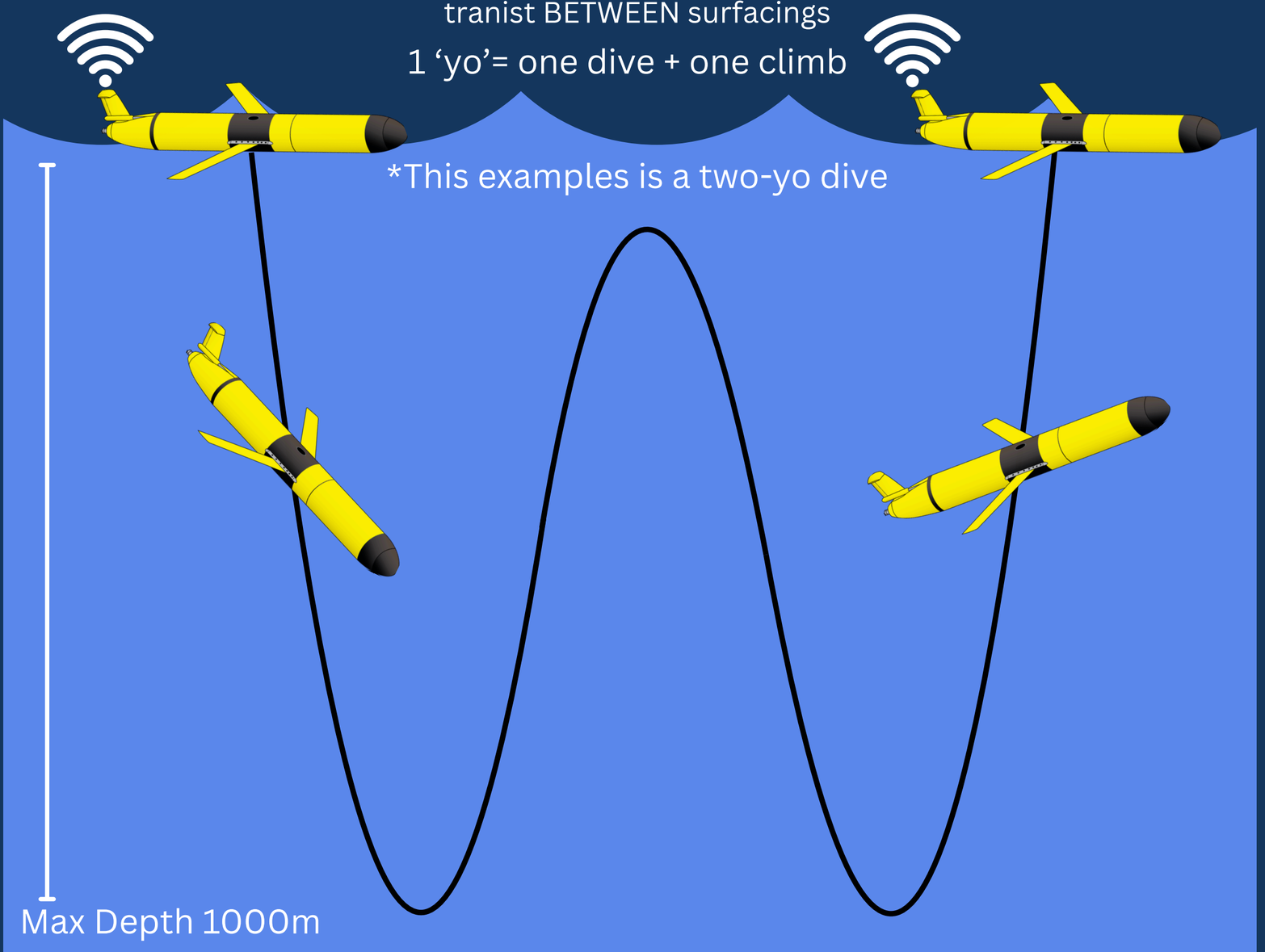
buoyancy = volume object * density medium

density = mass/volume

# High-Level Glider Transit

Segment = Portion of glider tranist BETWEEN surfacings

1 ‘yo’= one dive + one climb




<u>Buoyancy driven, where:</u>

buoyancy = volume object * density medium

density = mass/volume

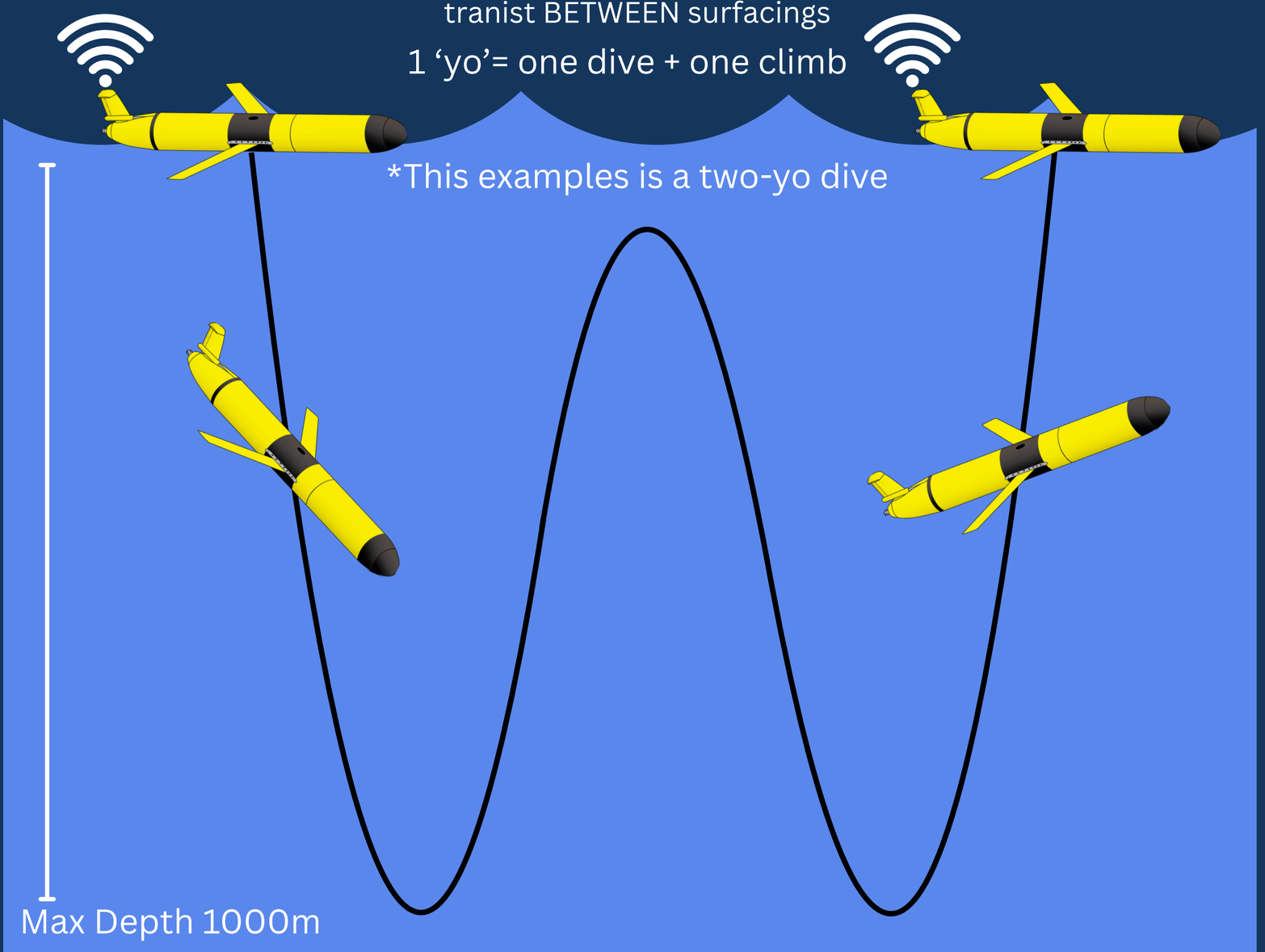
<u>Average horizontal speed:</u>

0.20 m/s; ~17 km/day

# High-Level Glider Transit

Segment = Portion of glider tranist BETWEEN surfacings

1 ‘yo’= one dive + one climb




<u>Buoyancy driven, where:</u>

buoyancy = volume object * density medium

density = mass/volume

<u>Average horizontal speed:</u>

0.20 m/s; ~17 km/day

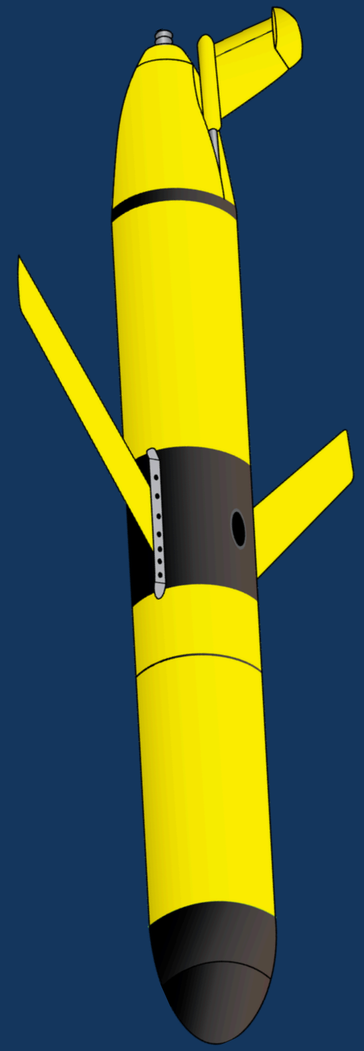
<u>Estimate surface interval:</u>

Surface interval = 2(Commanded depth (m)/10)

I.e., for a 900m dive:

2(900m/10) = 180 minutes

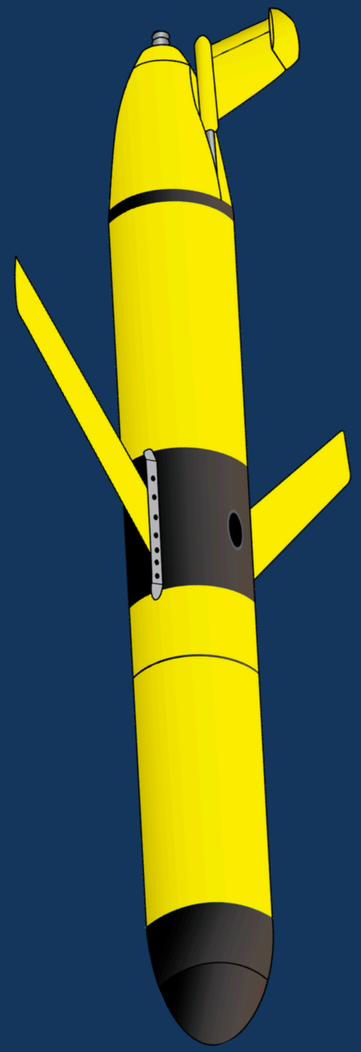
# High-Level Glider Comms



# High-Level Glider Comms

Dockserver with SFMC

Satellite: Iridium/GPS



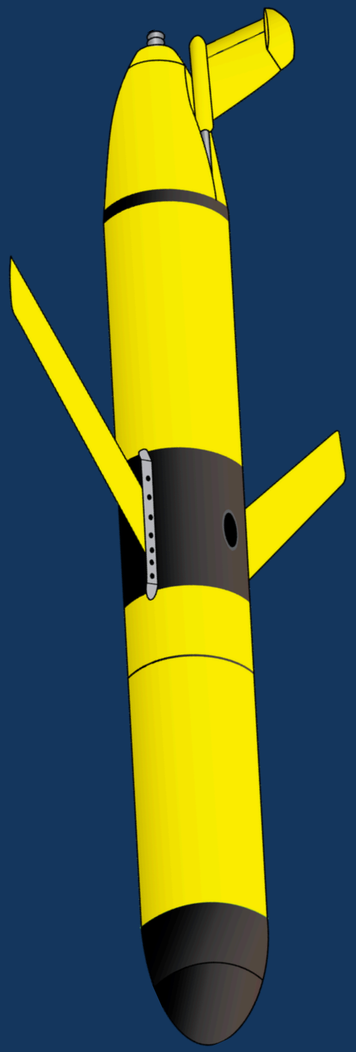
# High-Level Glider Comms

Dockserver with SFMC

Terminal Emulator Program

Satellite: Iridium/GPS

Freewave (wireless; line of sight)



# High-Level Glider Comms

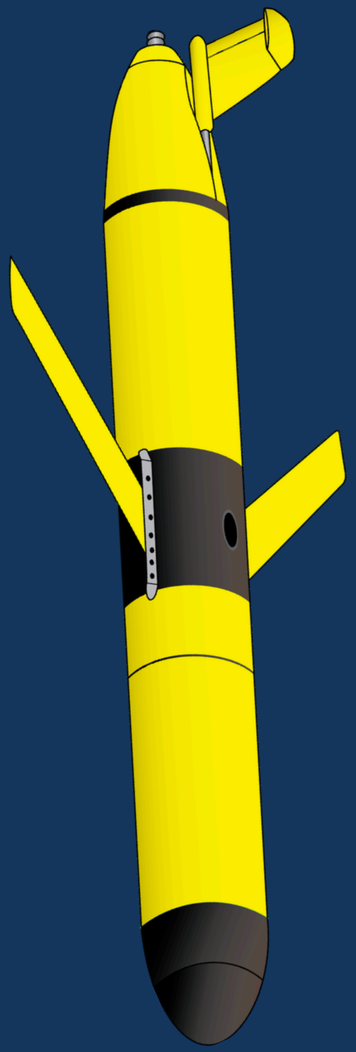
Dockserver with SFMC

Terminal Emulator Program

Satellite: Iridium/GPS

Freewave (wireless; line of sight)

Direct Comms (wired)



# High-Level Glider Comms

Dockserver with SFMC

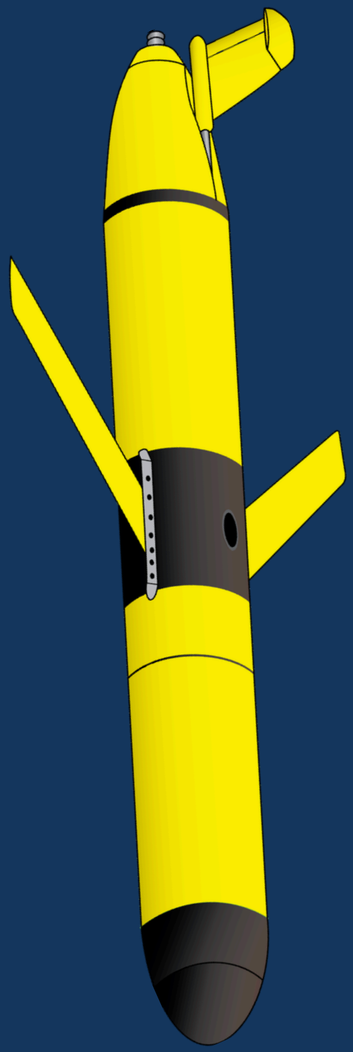
Terminal Emulator Program

ARGOS Backup GPS Website

Satellite: Iridium/GPS

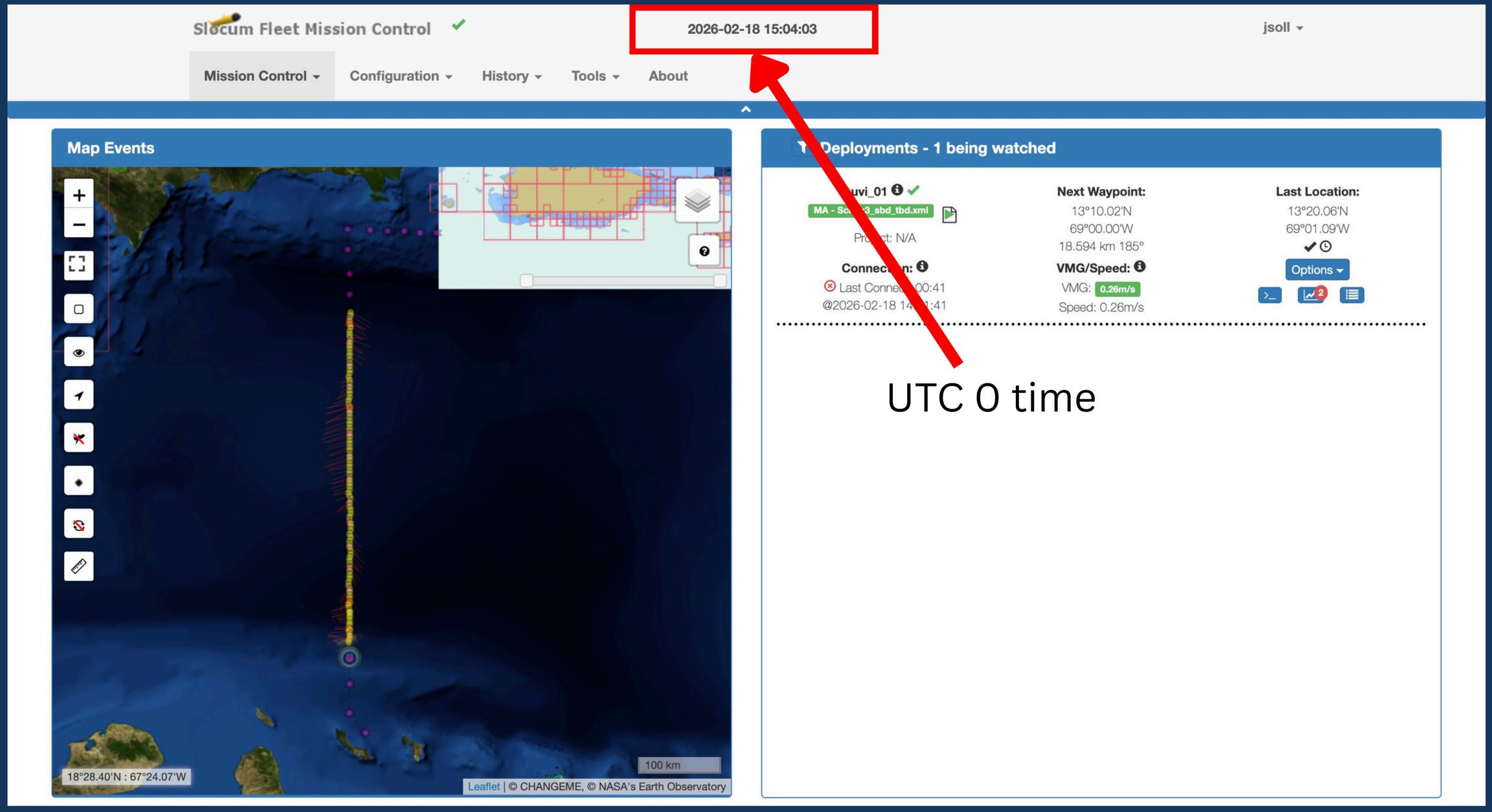
Freewave (wireless; line of sight)

Direct Comms (wired)

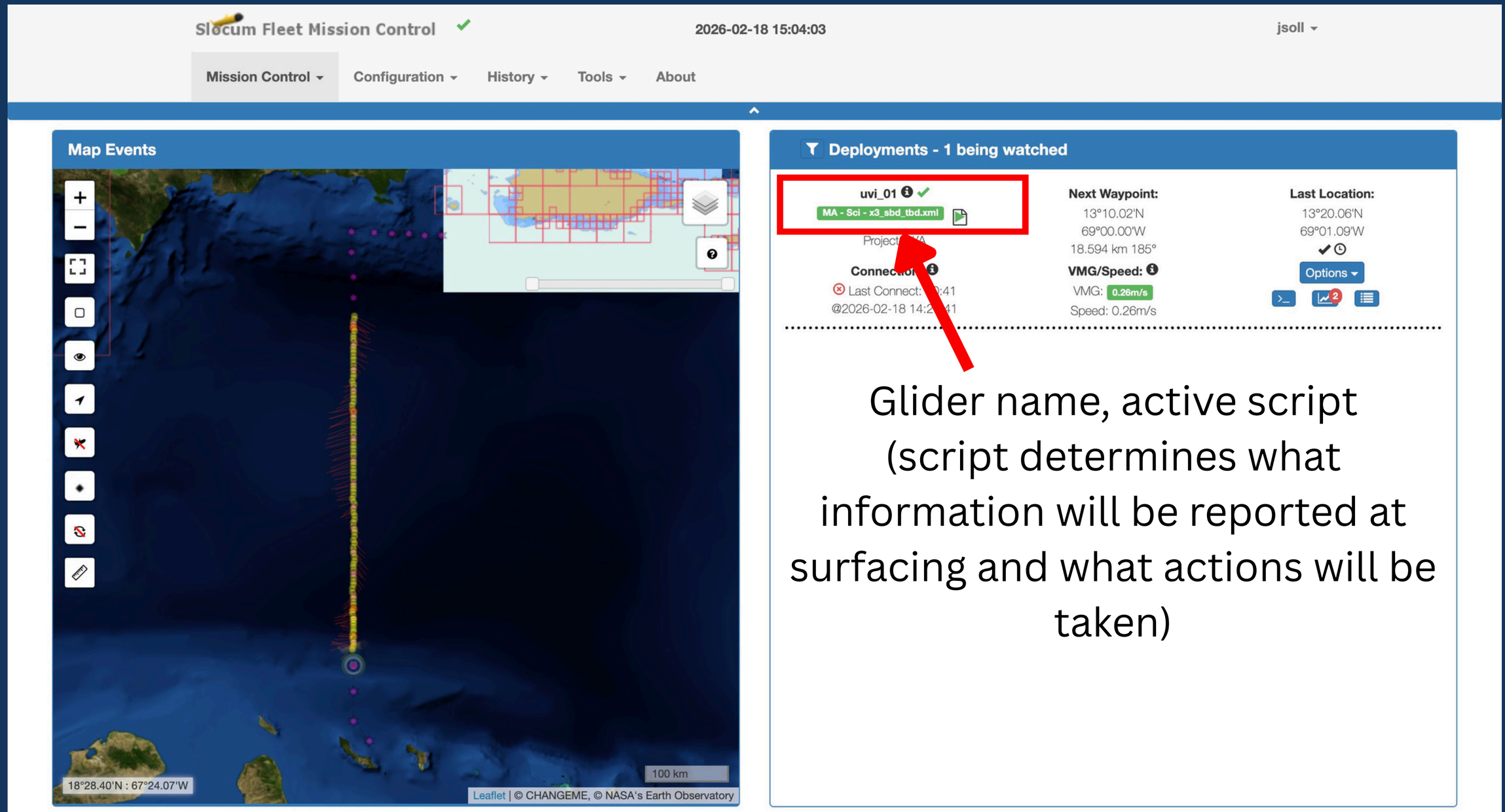


# Using Slocum Fleet Mission Control (SFMC): Dashboard

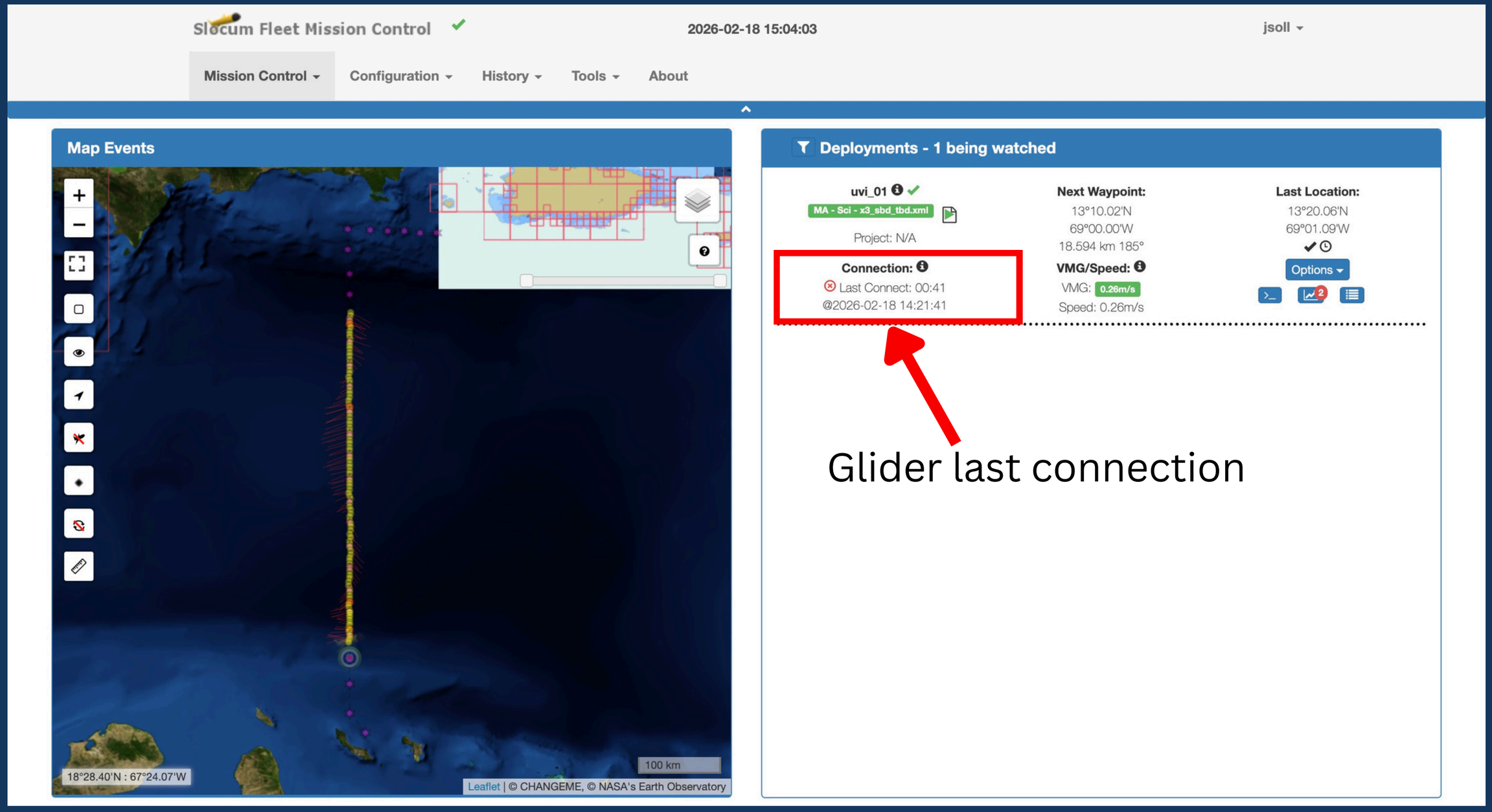
# SFMC Dashboard



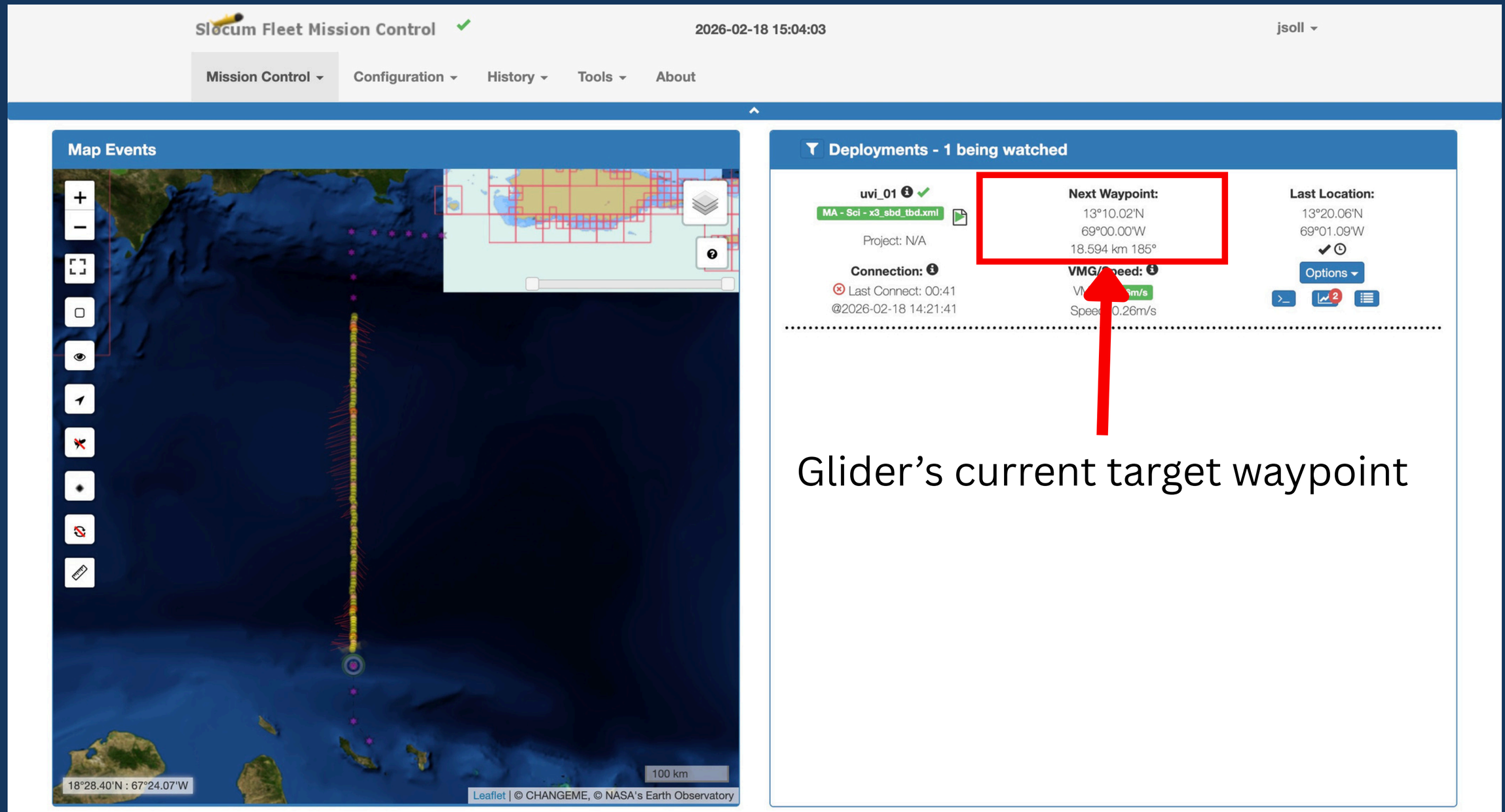
# SFMC Dashboard



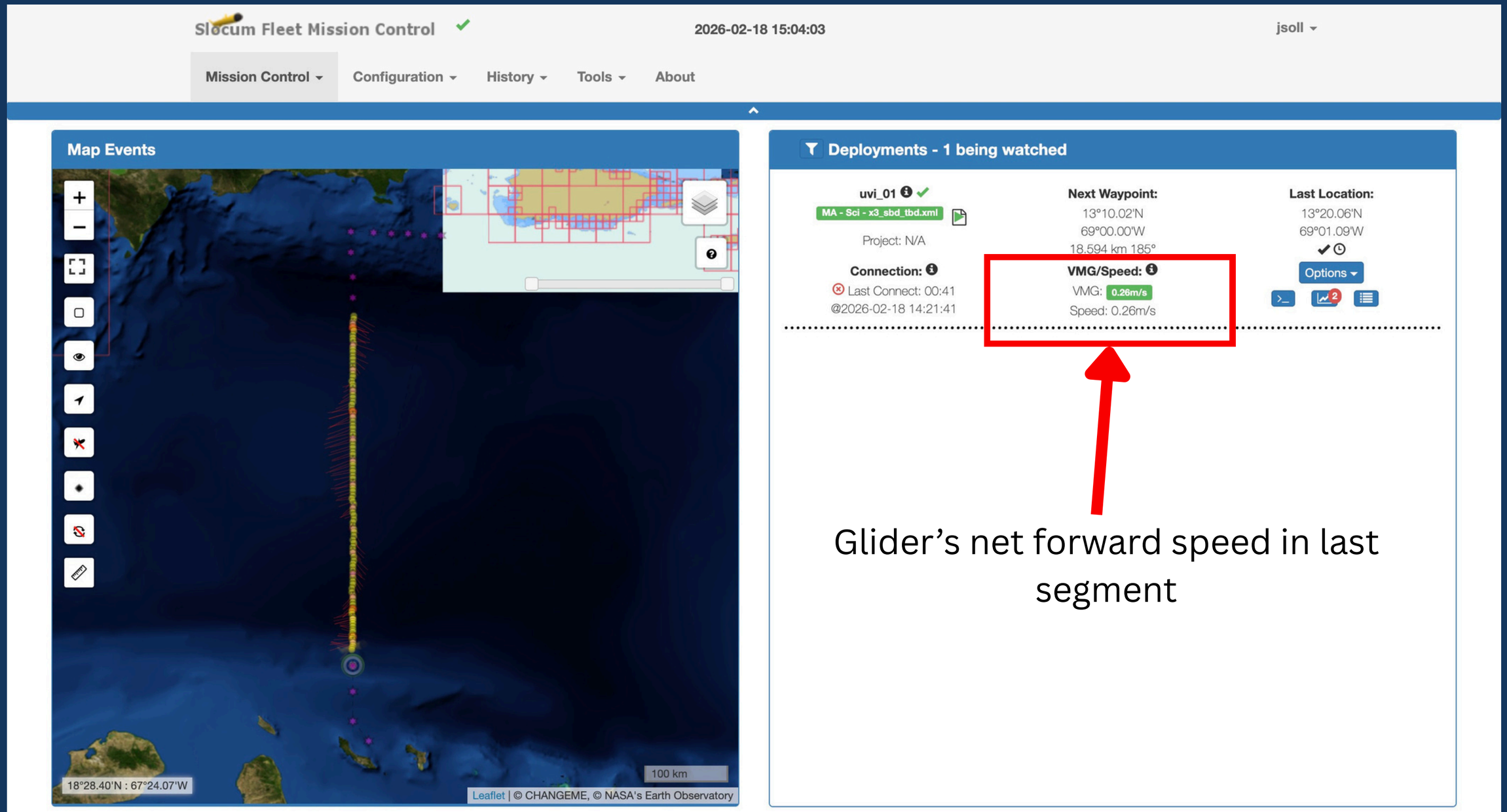
# SFMC Dashboard



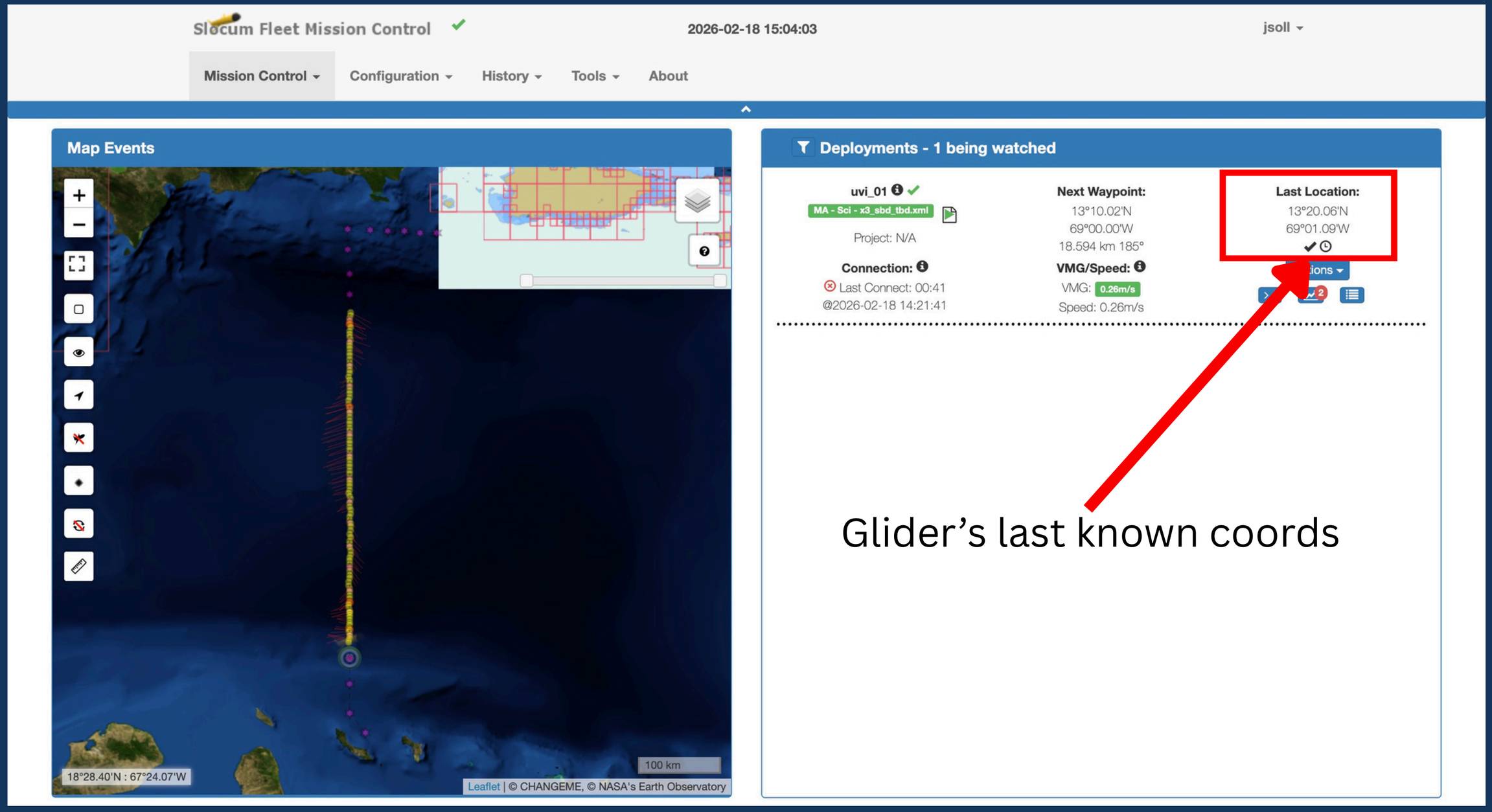
# SFMC Dashboard



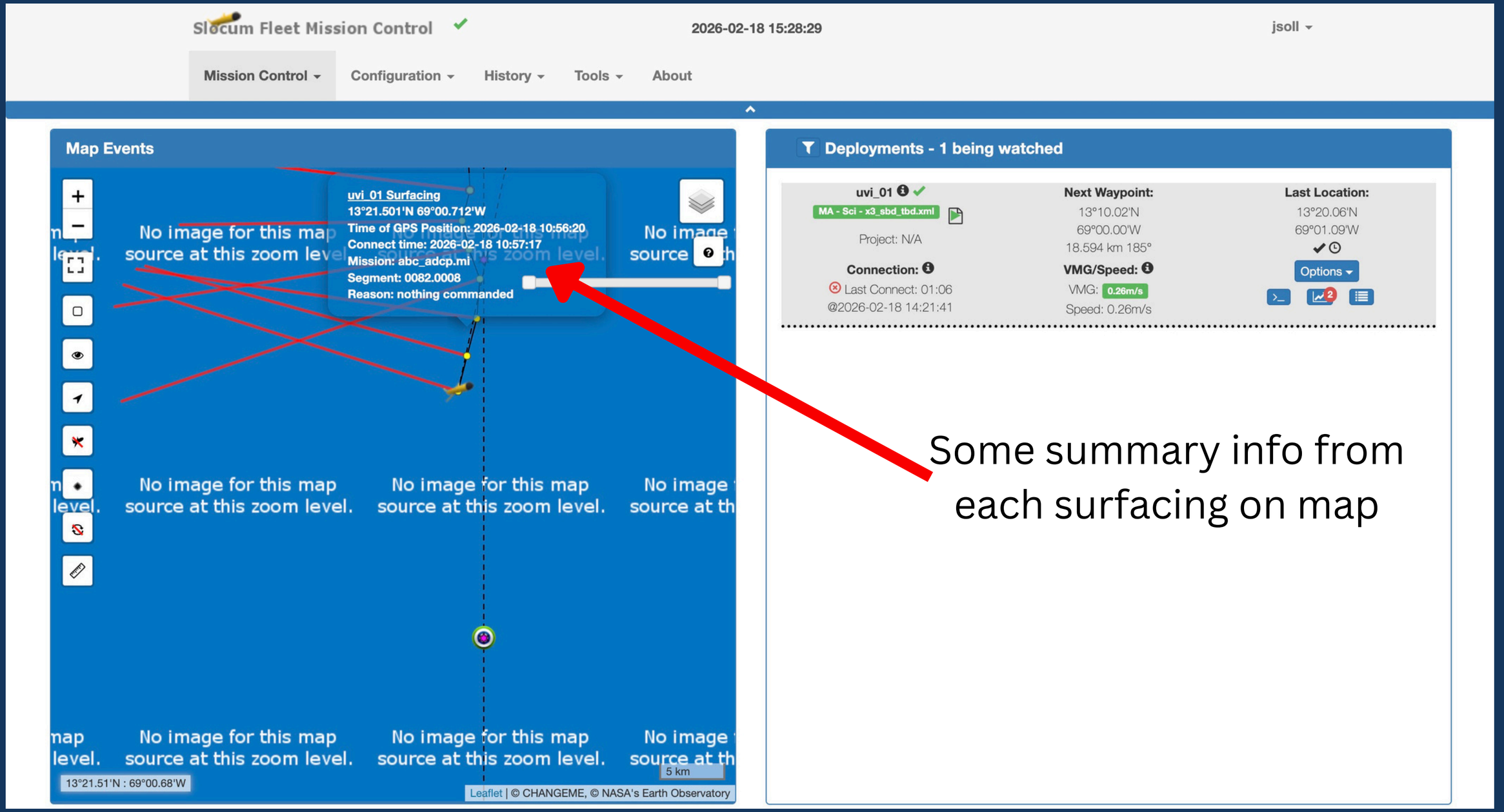
# SFMC Dashboard



# SFMC Dashboard

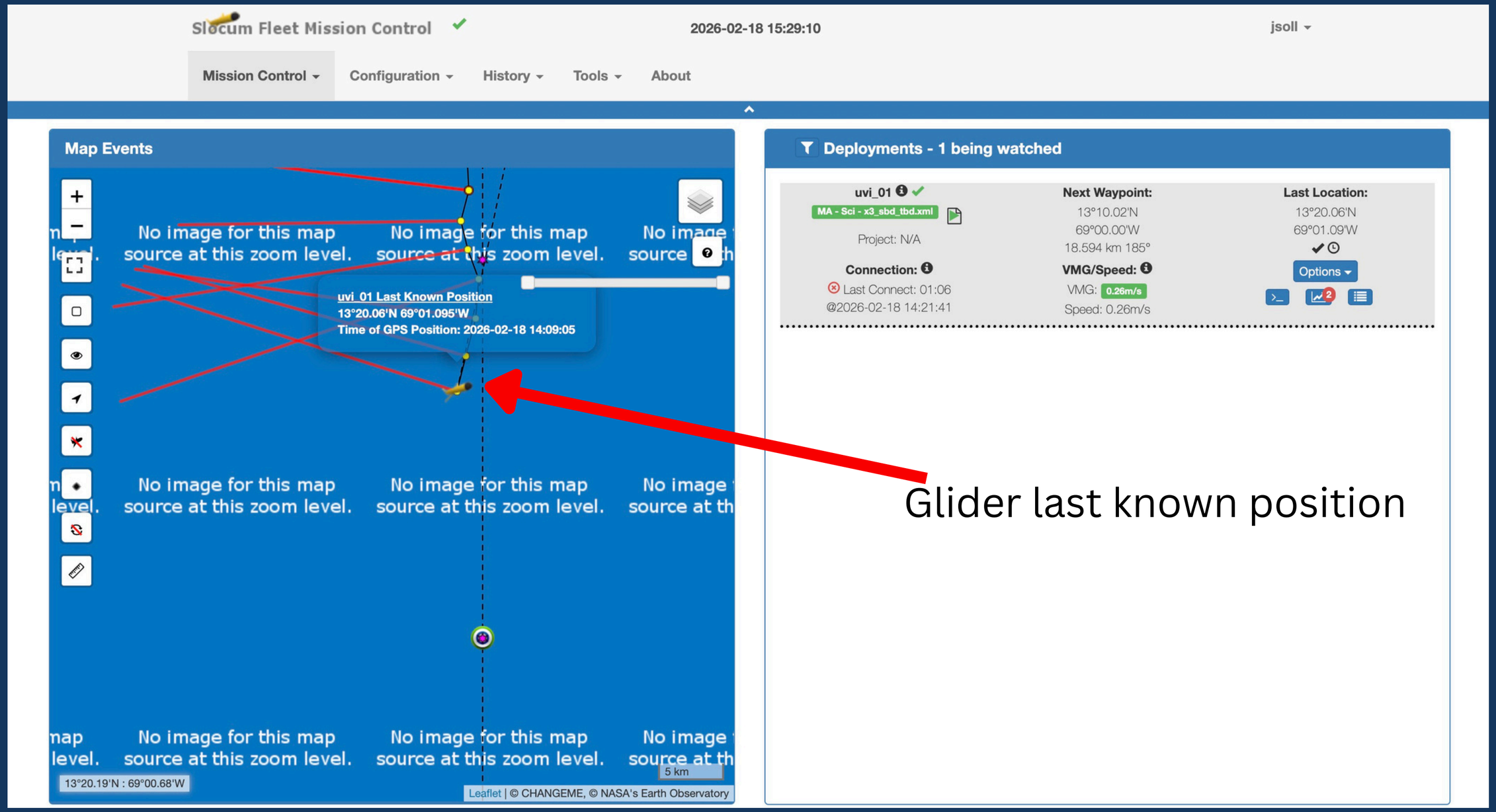


# SFMC Dashboard



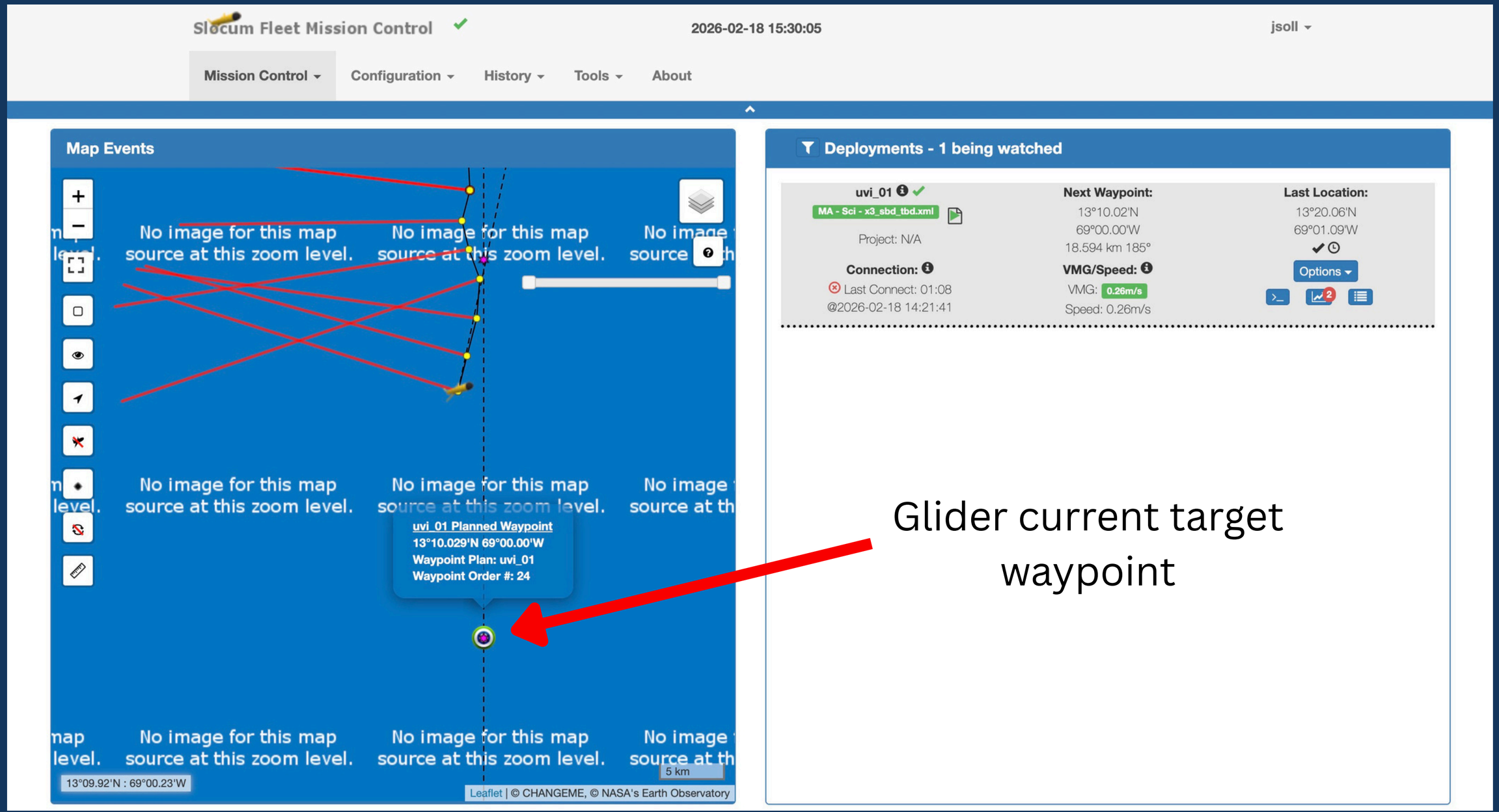

Some summary info from each surfacing on map

# SFMC Dashboard

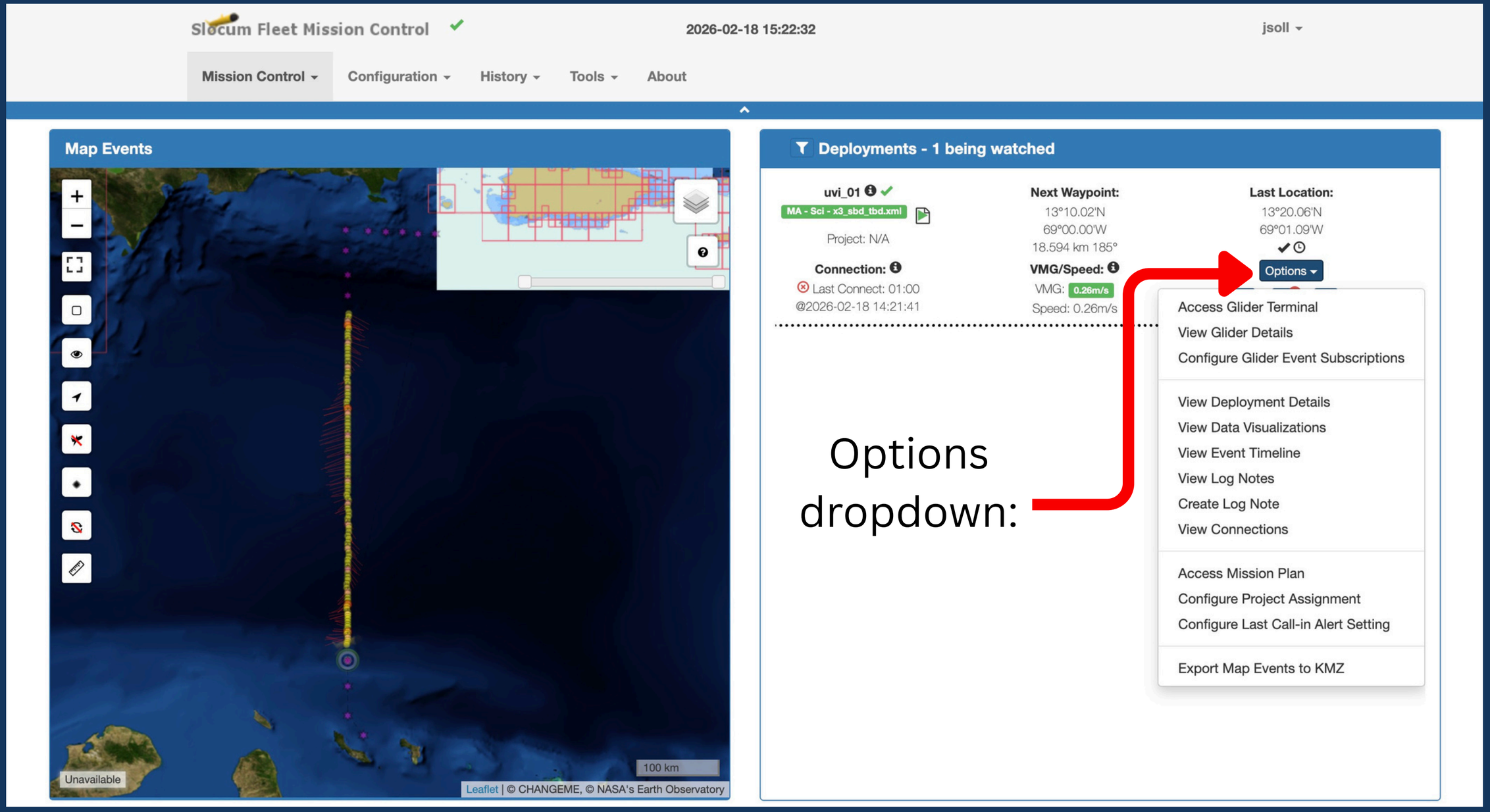



Glider last known position

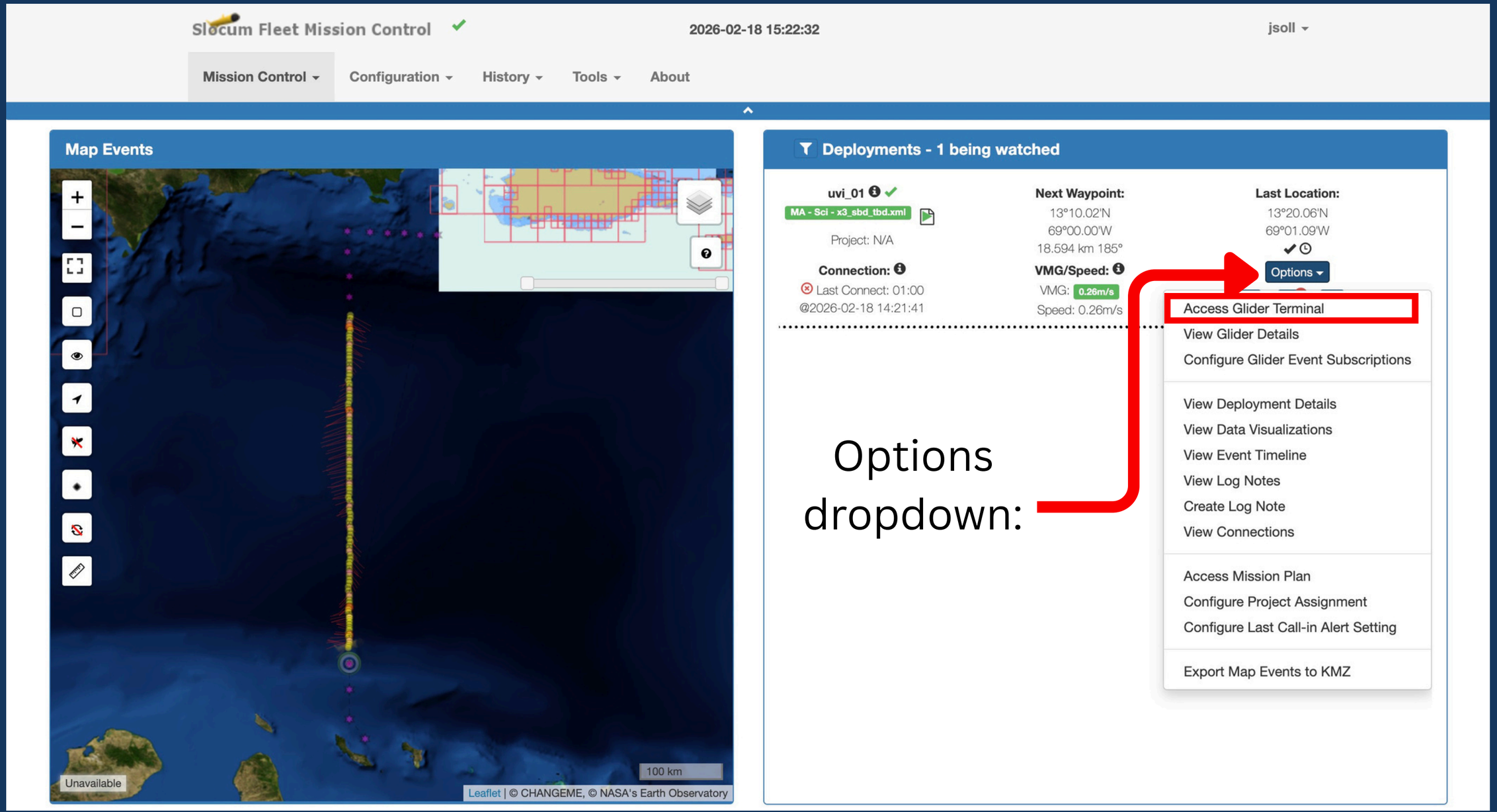
# SFMC Dashboard



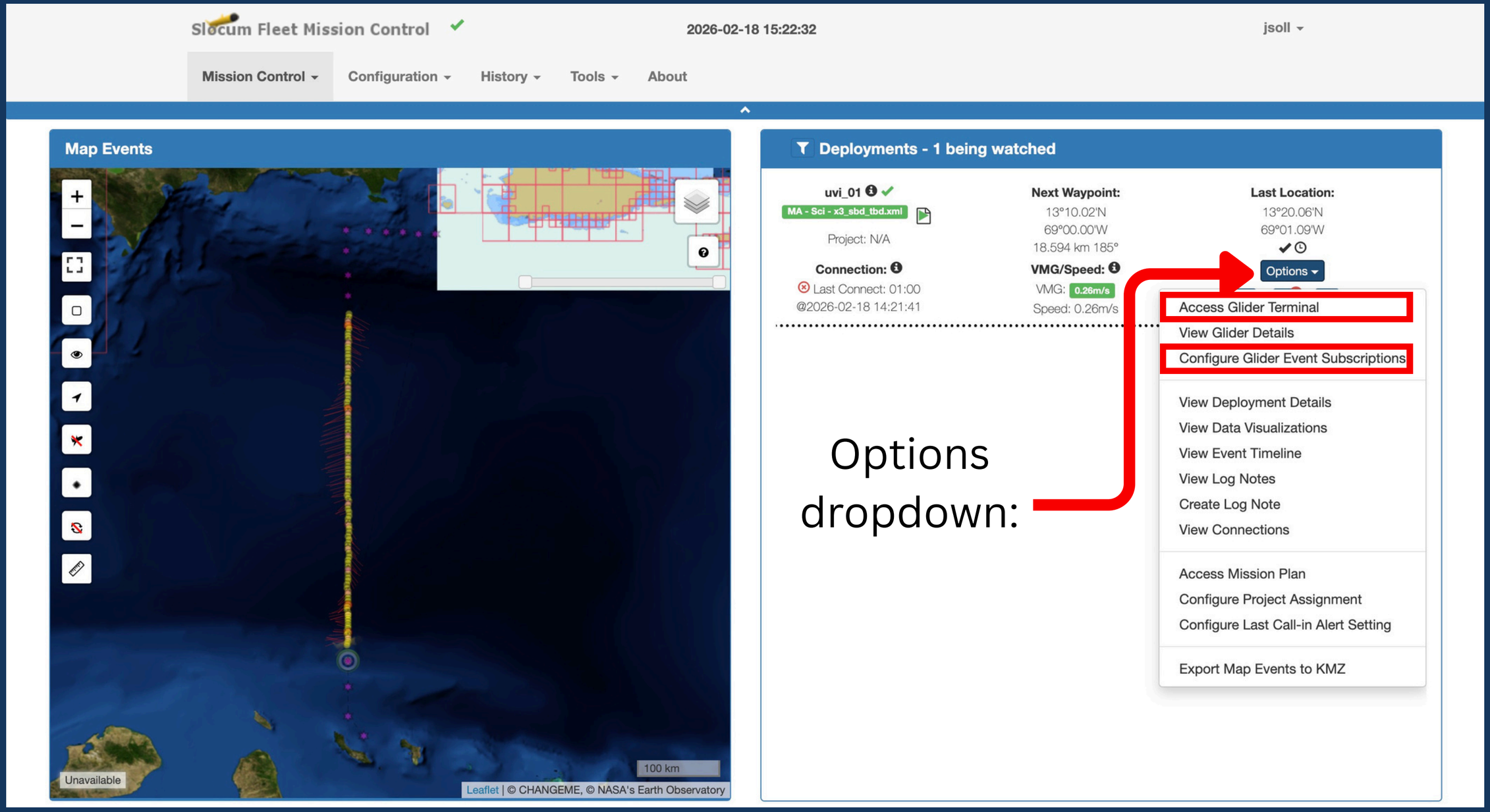
# SFMC Dashboard



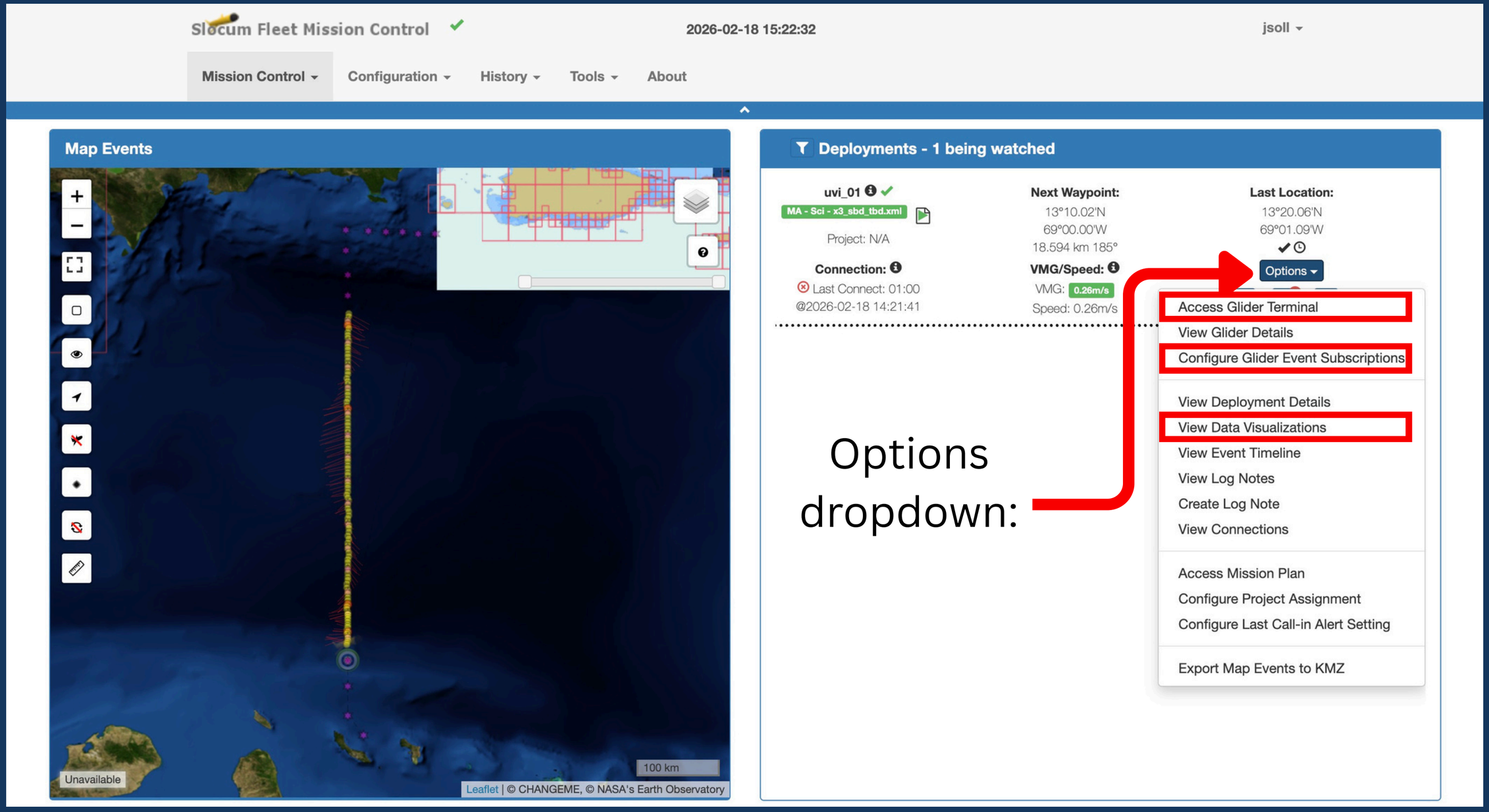
# SFMC Dashboard



# SFMC Dashboard

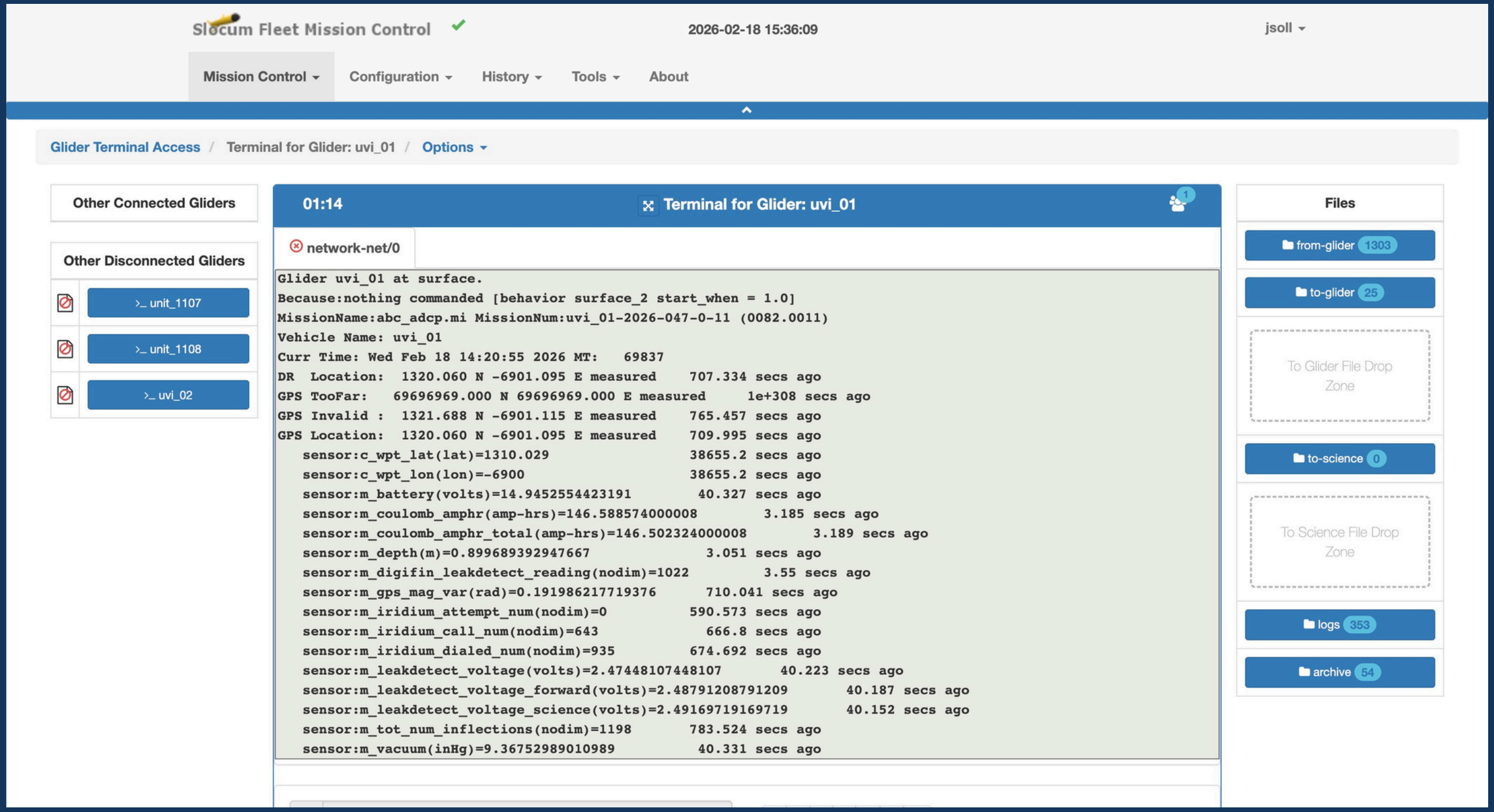


# SFMC Dashboard

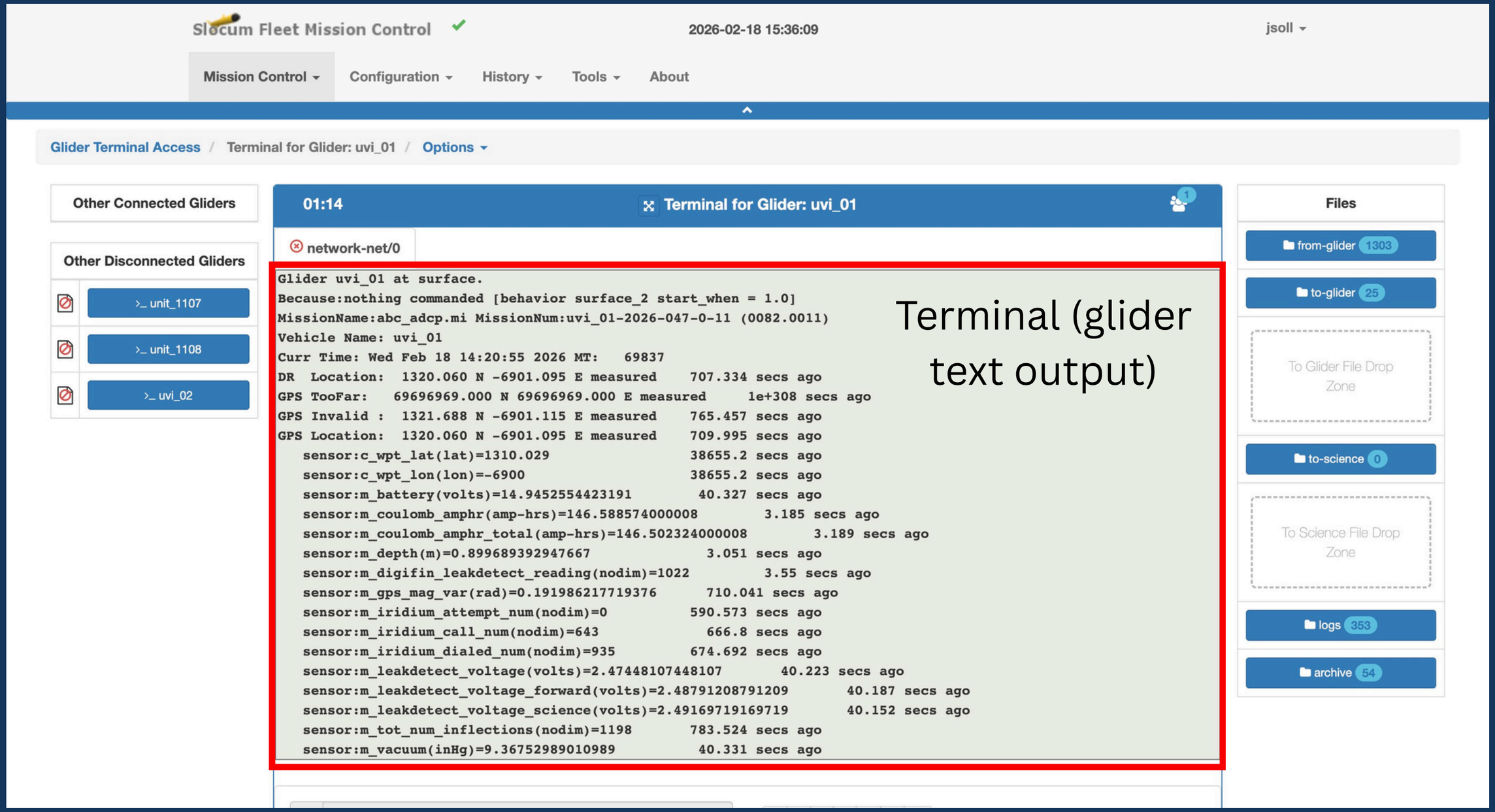


# Using Slocum Fleet Mission Control (SFMC): Terminal

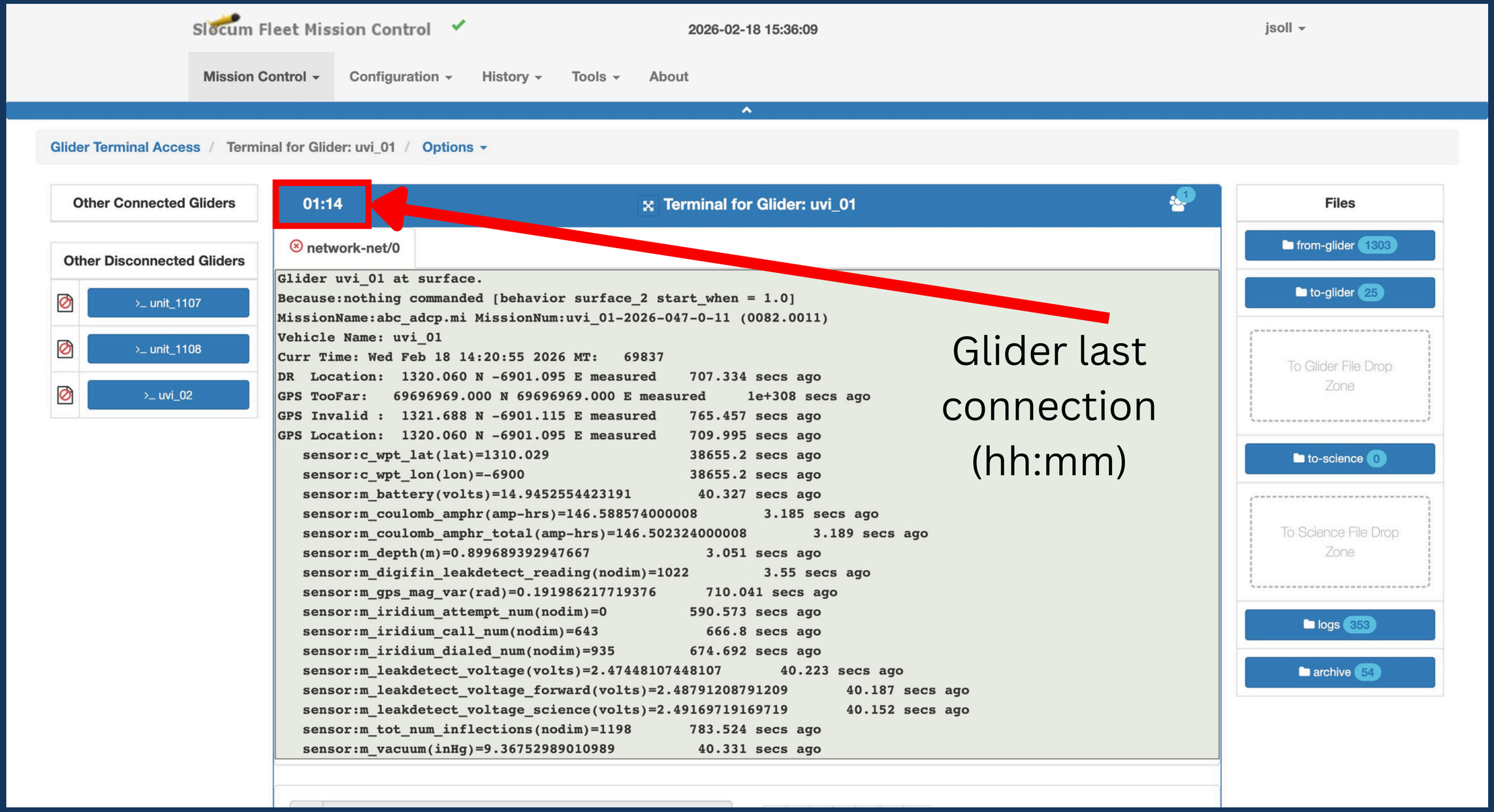
# SFMC Glider Terminal



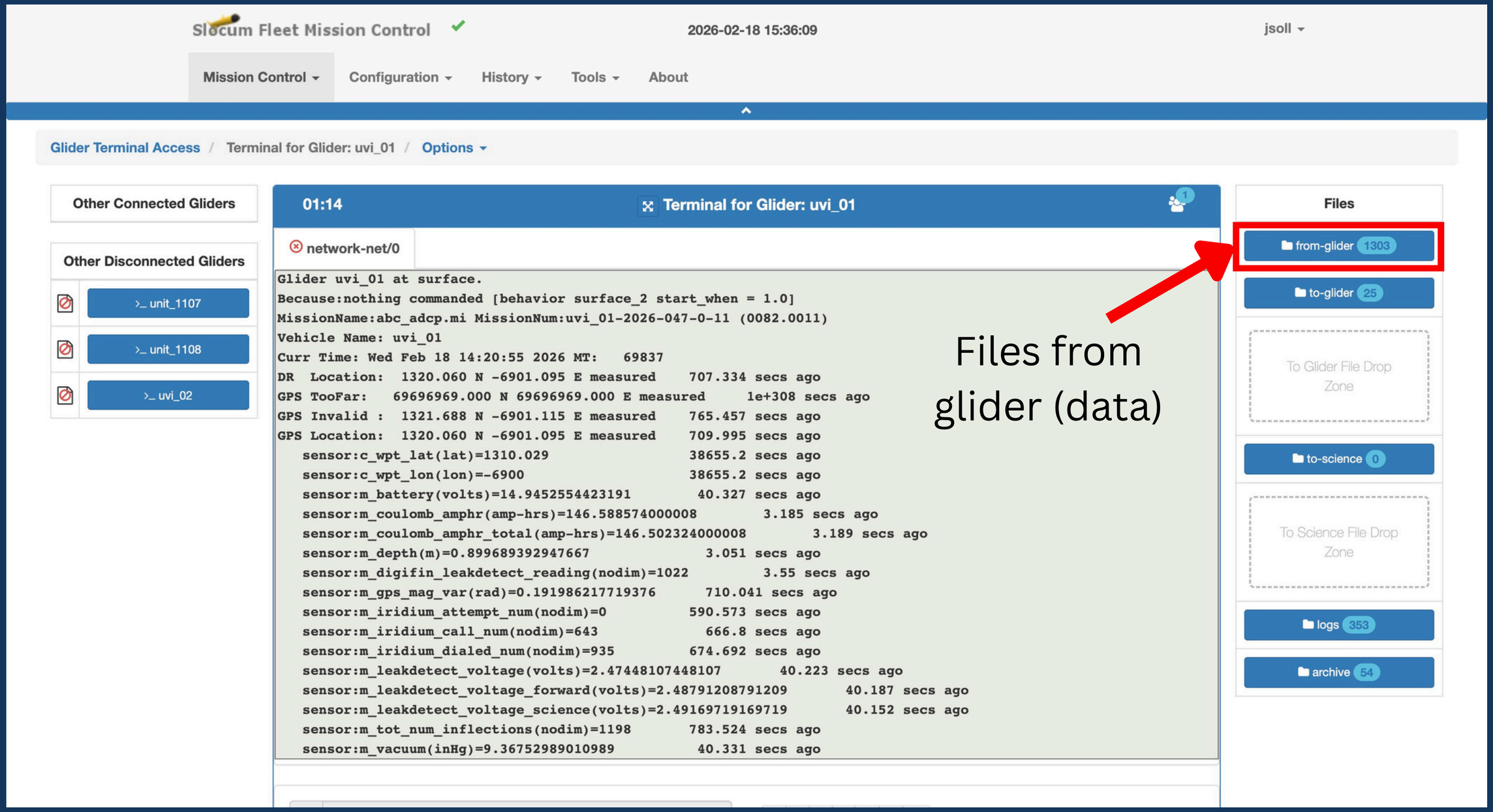
# SFMC Glider Terminal



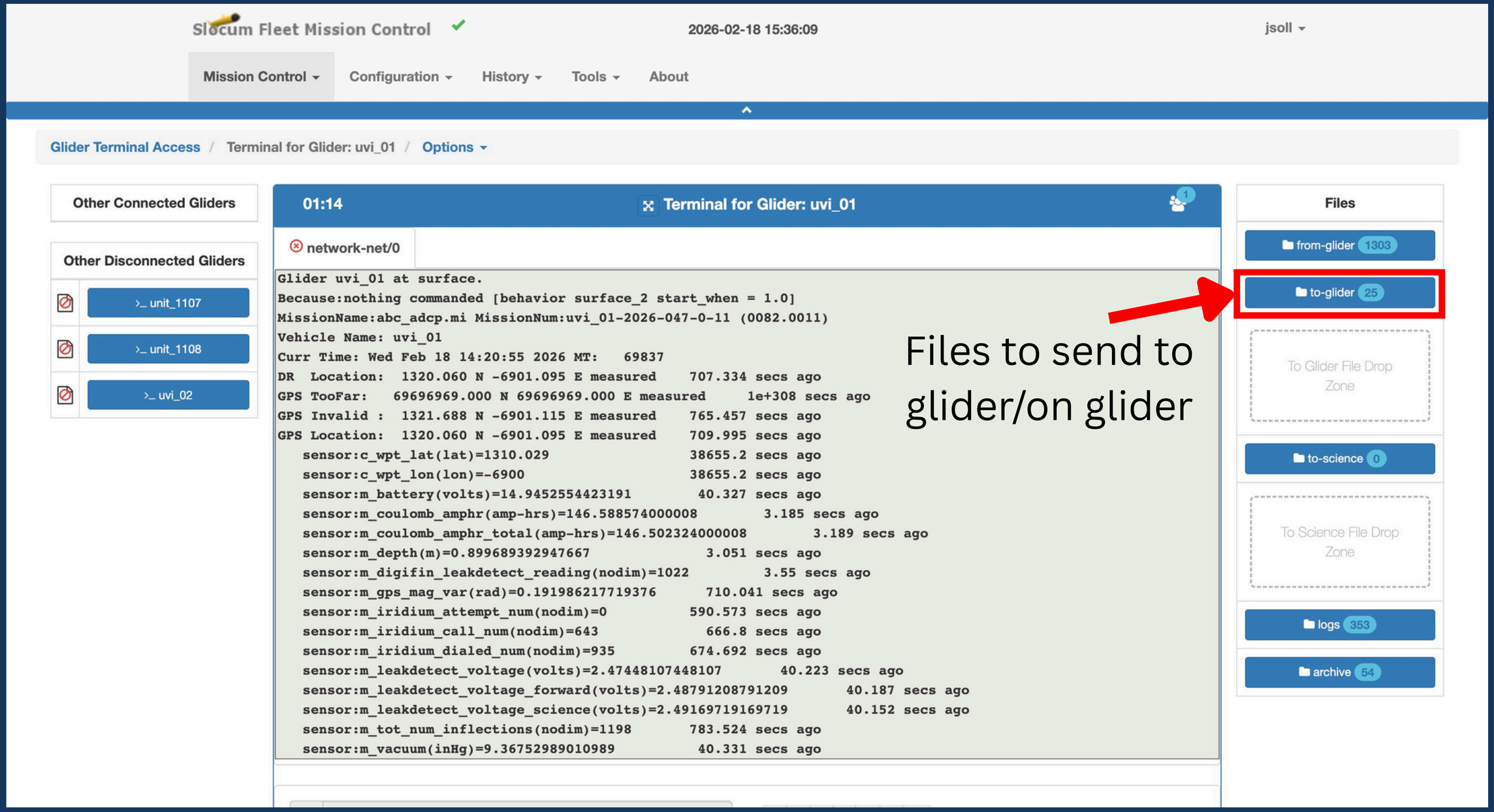
# SFMC Glider Terminal



# SFMC Glider Terminal

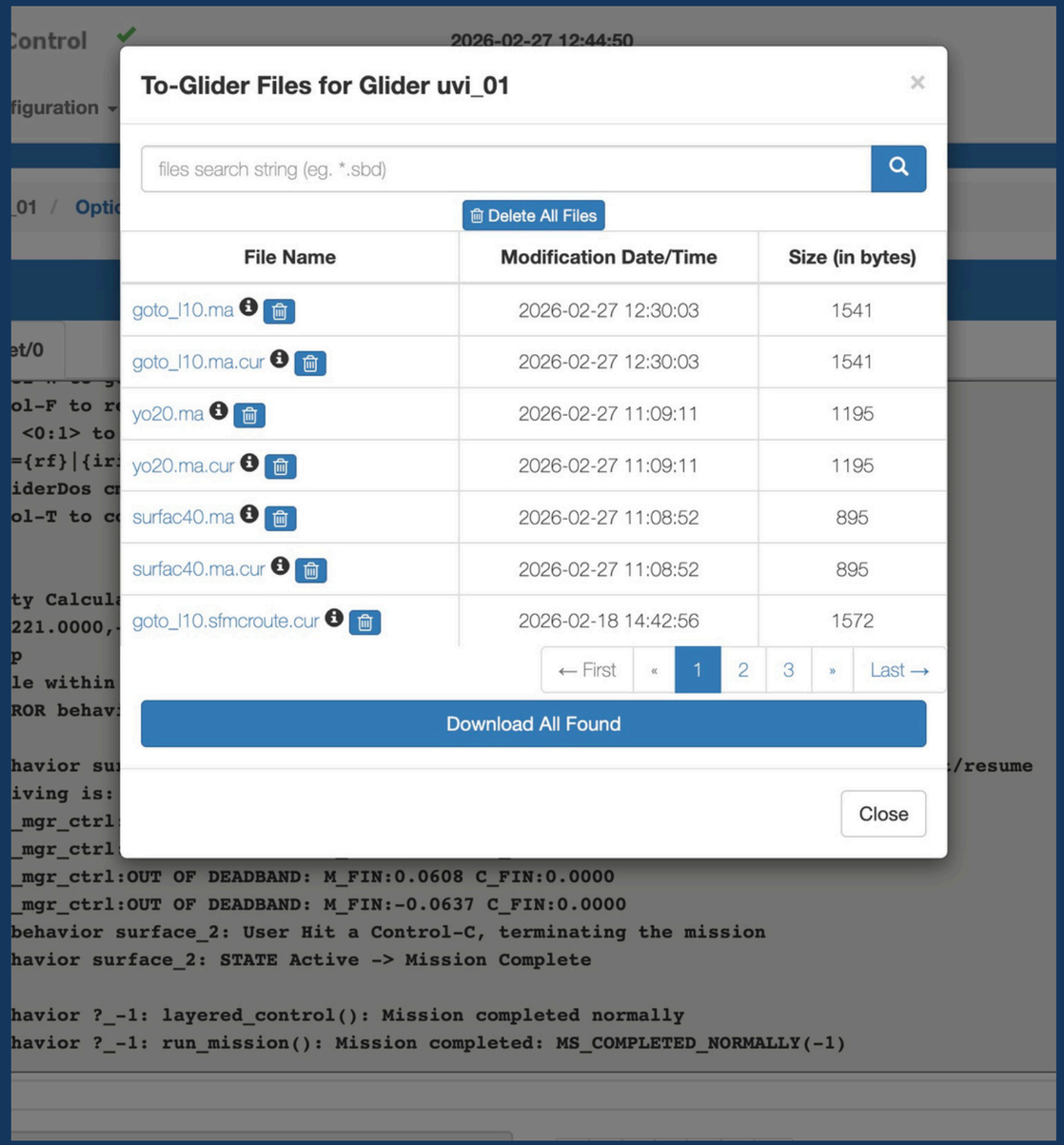


# SFMC Glider Terminal

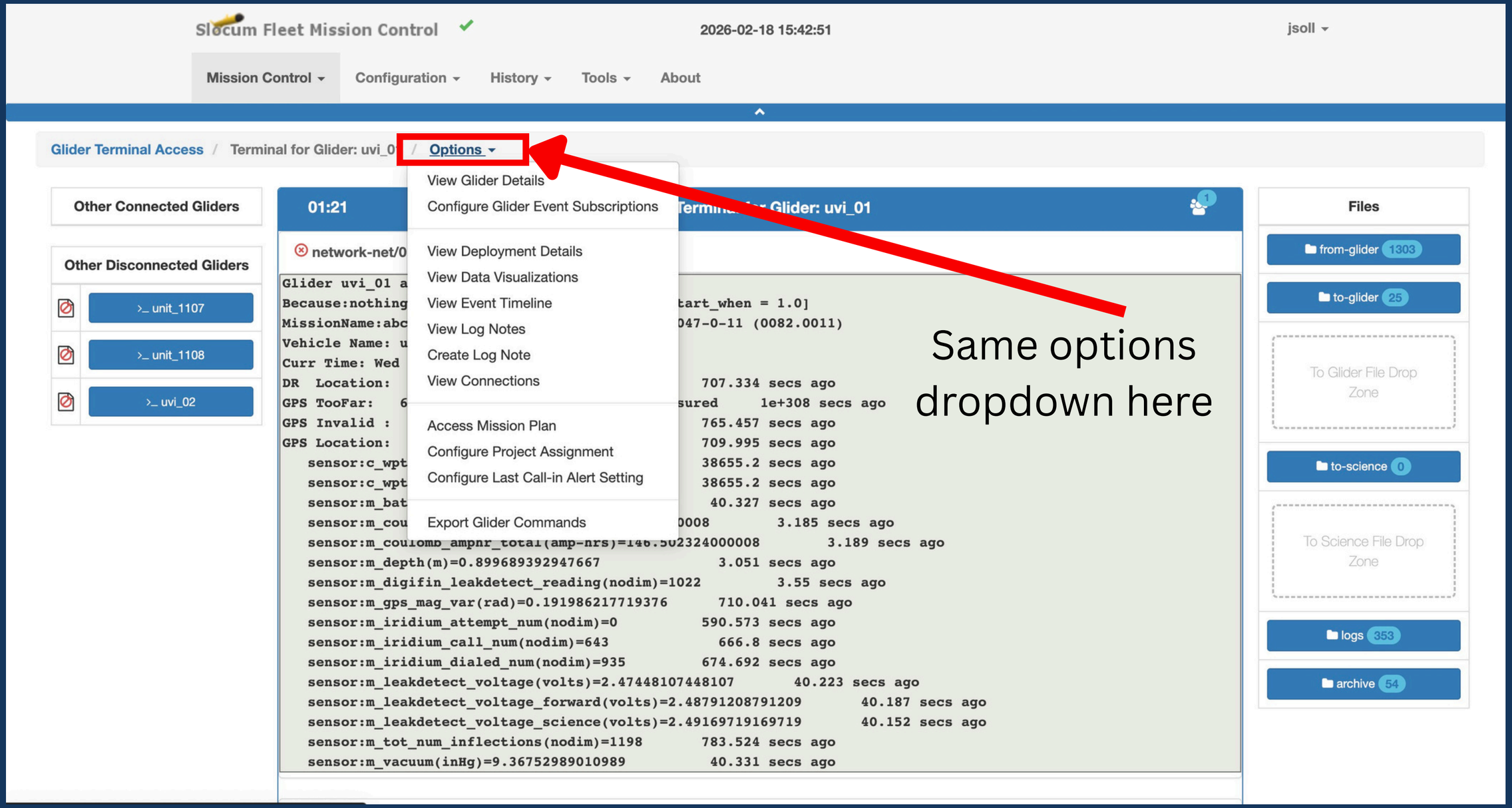


# SFMC Glider Terminal

- Files in the “To-Glider” folder are waiting for glider to surface to pull in those files and replace ones on the glider
- *.cur files are exact copies of other files
  - Not recognized by the glider, so not pulled onto glider
  - Serve as a record for pilot as to what files are currently on glider



# SFMC Glider Terminal



# Using Slocum Fleet Mission Control (SFMC): Terminal Header

# Reading the header

Goals

- Did the glider do what it was supposed to on the last segment?
- Does the glider look okay (from header output readings)?

```
Glider uvi_01 at surface.
Because:nothing commanded [behavior surface_2 start_when = 1.0]
MissionName:abc_adcp.mi MissionNum:uvi_01-2026-047-0-11 (0082.0011)
Vehicle Name: uvi_01
Curr Time: Wed Feb 18 14:20:55 2026 MT:   69837
DR  Location:  1320.060 N -6901.095 E measured    707.334 secs ago
GPS TooFar:   69696969.000 N 69696969.000 E measured     1e+308 secs ago
GPS Invalid :  1321.688 N -6901.115 E measured    765.457 secs ago
GPS Location:  1320.060 N -6901.095 E measured    709.995 secs ago
   sensor:c_wpt_lat(lat)=1310.029                  38655.2 secs ago
   sensor:c_wpt_lon(lon)=-6900                     38655.2 secs ago
   sensor:m_battery(volts)=14.9452554423191         40.327 secs ago
   sensor:m_coulomb_amphr(amp-hrs)=146.588574000008        3.185 secs ago
   sensor:m_coulomb_amphr_total(amp-hrs)=146.502324000008        3.189 secs ago
   sensor:m_depth(m)=0.899689392947667              3.051 secs ago
   sensor:m_digifin_leakdetect_reading(nodim)=1022         3.55 secs ago
   sensor:m_gps_mag_var(rad)=0.191986217719376      710.041 secs ago
   sensor:m_iridium_attempt_num(nodim)=0          590.573 secs ago
   sensor:m_iridium_call_num(nodim)=643             666.8 secs ago
   sensor:m_iridium_dialed_num(nodim)=935         674.692 secs ago
   sensor:m_leakdetect_voltage(volts)=2.47448107448107       40.223 secs ago
   sensor:m_leakdetect_voltage_forward(volts)=2.48791208791209       40.187 secs ago
   sensor:m_leakdetect_voltage_science(volts)=2.49169719169719       40.152 secs ago
   sensor:m_tot_num_inflections(nodim)=1198       783.524 secs ago
   sensor:m_vacuum(inHg)=9.36752989010989           40.331 secs ago
```

# Reading the header

- Sent from glider at each surfacing

```
Glider uvi_01 at surface.
Because:nothing commanded [behavior surface_2 start_when = 1.0]
MissionName:abc_adcp.mi MissionNum:uvi_01-2026-047-0-11 (0082.0011)
Vehicle Name: uvi_01
Curr Time: Wed Feb 18 14:20:55 2026 MT:   69837
DR  Location:  1320.060 N -6901.095 E measured    707.334 secs ago
GPS TooFar:   69696969.000 N 69696969.000 E measured     1e+308 secs ago
GPS Invalid :  1321.688 N -6901.115 E measured    765.457 secs ago
GPS Location:  1320.060 N -6901.095 E measured    709.995 secs ago
   sensor:c_wpt_lat(lat)=1310.029                   38655.2 secs ago
   sensor:c_wpt_lon(lon)=-6900                      38655.2 secs ago
   sensor:m_battery(volts)=14.9452554423191          40.327 secs ago
   sensor:m_coulomb_amphr(amp-hrs)=146.588574000008        3.185 secs ago
   sensor:m_coulomb_amphr_total(amp-hrs)=146.502324000008        3.189 secs ago
   sensor:m_depth(m)=0.899689392947667               3.051 secs ago
   sensor:m_digifin_leakdetect_reading(nodim)=1022         3.55 secs ago
   sensor:m_gps_mag_var(rad)=0.191986217719376      710.041 secs ago
   sensor:m_iridium_attempt_num(nodim)=0           590.573 secs ago
   sensor:m_iridium_call_num(nodim)=643              666.8 secs ago
   sensor:m_iridium_dialed_num(nodim)=935          674.692 secs ago
   sensor:m_leakdetect_voltage(volts)=2.47448107448107       40.223 secs ago
   sensor:m_leakdetect_voltage_forward(volts)=2.48791208791209       40.187 secs ago
   sensor:m_leakdetect_voltage_science(volts)=2.49169719169719       40.152 secs ago
   sensor:m_tot_num_inflections(nodim)=1198       783.524 secs ago
   sensor:m_vacuum(inHg)=9.36752989010989           40.331 secs ago
```

# Reading the header

- Sent from glider at each surfacing
- Contains important sensor readings related to glider condition

```
Glider uvi_01 at surface.
Because:nothing commanded [behavior surface_2 start_when = 1.0]
MissionName:abc_adcp.mi MissionNum:uvi_01-2026-047-0-11 (0082.0011)
Vehicle Name: uvi_01
Curr Time: Wed Feb 18 14:20:55 2026 MT:   69837
DR  Location:  1320.060 N -6901.095 E measured    707.334 secs ago
GPS TooFar:   69696969.000 N 69696969.000 E measured     1e+308 secs ago
GPS Invalid :  1321.688 N -6901.115 E measured    765.457 secs ago
GPS Location:  1320.060 N -6901.095 E measured    709.995 secs ago
   sensor:c_wpt_lat(lat)=1310.029                   38655.2 secs ago
   sensor:c_wpt_lon(lon)=-6900                      38655.2 secs ago
   sensor:m_battery(volts)=14.9452554423191          40.327 secs ago
   sensor:m_coulomb_amphr(amp-hrs)=146.588574000008        3.185 secs ago
   sensor:m_coulomb_amphr_total(amp-hrs)=146.502324000008        3.189 secs ago
   sensor:m_depth(m)=0.899689392947667               3.051 secs ago
   sensor:m_digifin_leakdetect_reading(nodim)=1022         3.55 secs ago
   sensor:m_gps_mag_var(rad)=0.191986217719376      710.041 secs ago
   sensor:m_iridium_attempt_num(nodim)=0          590.573 secs ago
   sensor:m_iridium_call_num(nodim)=643             666.8 secs ago
   sensor:m_iridium_dialed_num(nodim)=935         674.692 secs ago
   sensor:m_leakdetect_voltage(volts)=2.47448107448107       40.223 secs ago
   sensor:m_leakdetect_voltage_forward(volts)=2.48791208791209       40.187 secs ago
   sensor:m_leakdetect_voltage_science(volts)=2.49169719169719       40.152 secs ago
   sensor:m_tot_num_inflections(nodim)=1198       783.524 secs ago
   sensor:m_vacuum(inHg)=9.36752989010989          40.331 secs ago
```

# Reading the header

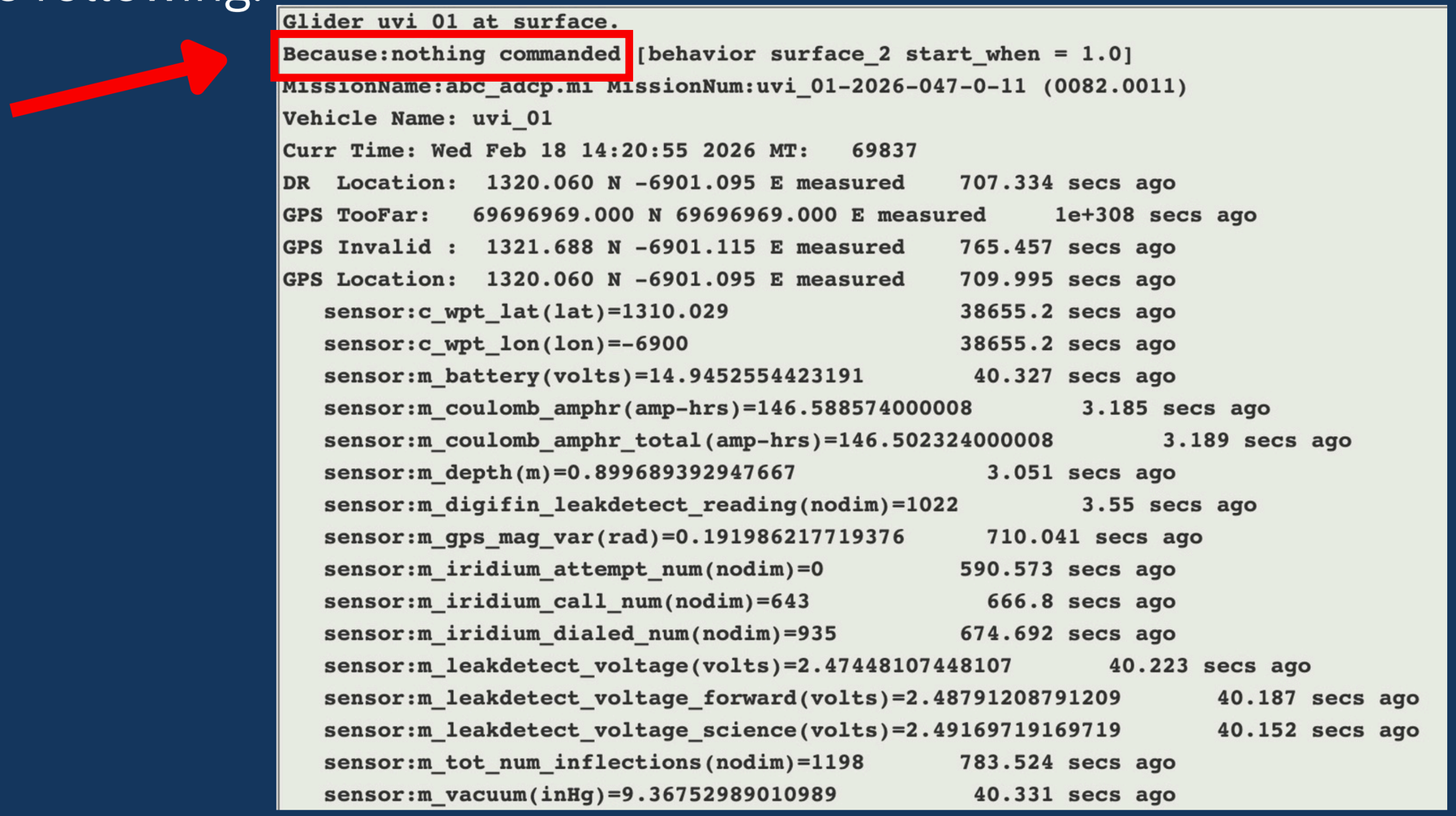
- Sent from glider at each surfacing
- Contains important sensor readings related to glider condition
- Important to keep track of glider's battery, vacuum, leak sensors

```
Glider uvi_01 at surface.
Because:nothing commanded [behavior surface_2 start_when = 1.0]
MissionName:abc_adcp.mi MissionNum:uvi_01-2026-047-0-11 (0082.0011)
Vehicle Name: uvi_01
Curr Time: Wed Feb 18 14:20:55 2026 MT:   69837
DR  Location:  1320.060 N -6901.095 E measured    707.334 secs ago
GPS TooFar:   69696969.000 N 69696969.000 E measured     1e+308 secs ago
GPS Invalid :  1321.688 N -6901.115 E measured    765.457 secs ago
GPS Location:  1320.060 N -6901.095 E measured    709.995 secs ago
   sensor:c_wpt_lat(lat)=1310.029                 38655.2 secs ago
   sensor:c_wpt_lon(lon)=-6900                    38655.2 secs ago
   sensor:m_battery(volts)=14.9452554423191        40.327 secs ago
   sensor:m_coulomb_amphr(amp-hrs)=146.588574000008        3.185 secs ago
   sensor:m_coulomb_amphr_total(amp-hrs)=146.502324000008        3.189 secs ago
   sensor:m_depth(m)=0.899689392947667              3.051 secs ago
   sensor:m_digifin_leakdetect_reading(nodim)=1022         3.55 secs ago
   sensor:m_gps_mag_var(rad)=0.191986217719376      710.041 secs ago
   sensor:m_iridium_attempt_num(nodim)=0          590.573 secs ago
   sensor:m_iridium_call_num(nodim)=643             666.8 secs ago
   sensor:m_iridium_dialed_num(nodim)=935         674.692 secs ago
   sensor:m_leakdetect_voltage(volts)=2.47448107448107       40.223 secs ago
   sensor:m_leakdetect_voltage_forward(volts)=2.48791208791209       40.187 secs ago
   sensor:m_leakdetect_voltage_science(volts)=2.49169719169719       40.152 secs ago
   sensor:m_tot_num_inflections(nodim)=1198       783.524 secs ago
   sensor:m_vacuum(inHg)=9.36752989010989          40.331 secs ago
```

# Reading the header

For UVI our most important reading to inspect are currently the following:

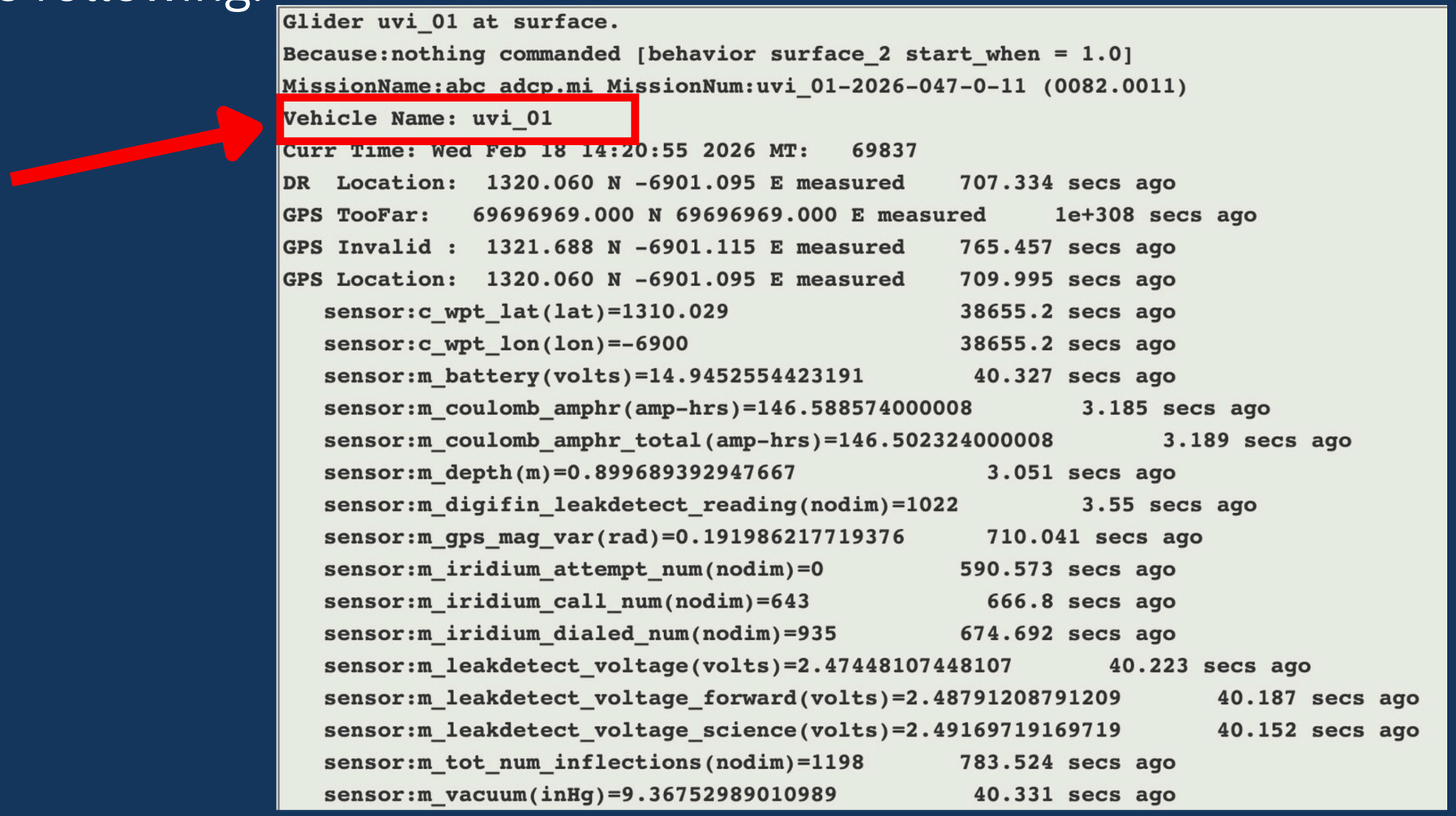
- Why did it surface



# Reading the header

For UVI our most important reading to inspect are currently the following:

- Why did it surface
- Which glider is this?



```
Glider uvi_01 at surface.
Because:nothing commanded [behavior surface_2 start_when = 1.0]
MissionName:abc_adcp.mi MissionNum:uvi_01-2026-047-0-11 (0082.0011)
Vehicle Name: uvi_01
Curr Time: Wed Feb 18 14:20:55 2026 MT:   69837
DR  Location:  1320.060 N -6901.095 E measured    707.334 secs ago
GPS TooFar:   69696969.000 N 69696969.000 E measured     1e+308 secs ago
GPS Invalid :  1321.688 N -6901.115 E measured    765.457 secs ago
GPS Location:  1320.060 N -6901.095 E measured    709.995 secs ago
   sensor:c_wpt_lat(lat)=1310.029                  38655.2 secs ago
   sensor:c_wpt_lon(lon)=-6900                     38655.2 secs ago
   sensor:m_battery(volts)=14.9452554423191         40.327 secs ago
   sensor:m_coulomb_amphr(amp-hrs)=146.588574000008        3.185 secs ago
   sensor:m_coulomb_amphr_total(amp-hrs)=146.502324000008        3.189 secs ago
   sensor:m_depth(m)=0.899689392947667               3.051 secs ago
   sensor:m_digifin_leakdetect_reading(nodim)=1022         3.55 secs ago
   sensor:m_gps_mag_var(rad)=0.191986217719376      710.041 secs ago
   sensor:m_iridium_attempt_num(nodim)=0           590.573 secs ago
   sensor:m_iridium_call_num(nodim)=643              666.8 secs ago
   sensor:m_iridium_dialed_num(nodim)=935          674.692 secs ago
   sensor:m_leakdetect_voltage(volts)=2.47448107448107       40.223 secs ago
   sensor:m_leakdetect_voltage_forward(volts)=2.48791208791209       40.187 secs ago
   sensor:m_leakdetect_voltage_science(volts)=2.49169719169719       40.152 secs ago
   sensor:m_tot_num_inflections(nodim)=1198        783.524 secs ago
   sensor:m_vacuum(inHg)=9.36752989010989           40.331 secs ago
```

# Reading the header

For UVI our most important reading to inspect are currently the following:

- Why did it surface
- Which glider is this?
- Where is the glider?

```
Glider uvi_01 at surface.
Because:nothing commanded [behavior surface_2 start_when = 1.0]
MissionName:abc_adcp.mi MissionNum:uvi_01-2026-047-0-11 (0082.0011)
Vehicle Name: uvi_01
Curr Time: Wed Feb 18 14:20:55 2026 MT:   69837
DR  Location:  1320.060 N -6901.095 E measured    707.334 secs ago
GPS TooFar:   69696969.000 N 69696969.000 E measured     1e+308 secs ago
GPS Invalid :  1321.688 N -6901.115 E measured    765.457 secs ago
GPS Location:  1320.060 N -6901.095 E measured    709.995 secs ago
   sensor:c_wpt_lat(lat)=1310.029                  38655.2 secs ago
   sensor:c_wpt_lon(lon)=-6900                     38655.2 secs ago
   sensor:m_battery(volts)=14.9452554423191         40.327 secs ago
   sensor:m_coulomb_amphr(amp-hrs)=146.588574000008        3.185 secs ago
   sensor:m_coulomb_amphr_total(amp-hrs)=146.502324000008        3.189 secs ago
   sensor:m_depth(m)=0.899689392947667               3.051 secs ago
   sensor:m_digifin_leakdetect_reading(nodim)=1022         3.55 secs ago
   sensor:m_gps_mag_var(rad)=0.191986217719376      710.041 secs ago
   sensor:m_iridium_attempt_num(nodim)=0           590.573 secs ago
   sensor:m_iridium_call_num(nodim)=643              666.8 secs ago
   sensor:m_iridium_dialed_num(nodim)=935          674.692 secs ago
   sensor:m_leakdetect_voltage(volts)=2.47448107448107       40.223 secs ago
   sensor:m_leakdetect_voltage_forward(volts)=2.48791208791209       40.187 secs ago
   sensor:m_leakdetect_voltage_science(volts)=2.49169719169719       40.152 secs ago
   sensor:m_tot_num_inflections(nodim)=1198        783.524 secs ago
   sensor:m_vacuum(inHg)=9.36752989010989           40.331 secs ago
```

# Reading the header

For UVI our most important reading to inspect are currently the following:

- Why did it surface
- Which glider is this?
- Where is the glider?
- What is the battery voltage?

```
Glider uvi_01 at surface.
Because:nothing commanded [behavior surface_2 start_when = 1.0]
MissionName:abc_adcp.mi MissionNum:uvi_01-2026-047-0-11 (0082.0011)
Vehicle Name: uvi_01
Curr Time: Wed Feb 18 14:20:55 2026 MT:   69837
DR  Location:  1320.060 N -6901.095 E measured    707.334 secs ago
GPS TooFar:   69696969.000 N 69696969.000 E measured     1e+308 secs ago
GPS Invalid :  1321.688 N -6901.115 E measured    765.457 secs ago
GPS Location:  1320.060 N -6901.095 E measured    709.995 secs ago
   sensor:c_wpt_lat(lat)=1310.029                 38655.2 secs ago
   sensor:c_wpt_lon(lon)=-6900                    38655.2 secs ago
   sensor:m_battery(volts)=14.9452554423191        40.327 secs ago
   sensor:m_coulomb_amphr(amp-hrs)=146.588574000008        3.185 secs ago
   sensor:m_coulomb_amphr_total(amp-hrs)=146.502324000008        3.189 secs ago
   sensor:m_depth(m)=0.899689392947667              3.051 secs ago
   sensor:m_digifin_leakdetect_reading(nodim)=1022         3.55 secs ago
   sensor:m_gps_mag_var(rad)=0.191986217719376      710.041 secs ago
   sensor:m_iridium_attempt_num(nodim)=0          590.573 secs ago
   sensor:m_iridium_call_num(nodim)=643             666.8 secs ago
   sensor:m_iridium_dialed_num(nodim)=935         674.692 secs ago
   sensor:m_leakdetect_voltage(volts)=2.47448107448107       40.223 secs ago
   sensor:m_leakdetect_voltage_forward(volts)=2.48791208791209       40.187 secs ago
   sensor:m_leakdetect_voltage_science(volts)=2.49169719169719       40.152 secs ago
   sensor:m_tot_num_inflections(nodim)=1198       783.524 secs ago
   sensor:m_vacuum(inHg)=9.36752989010989          40.331 secs ago
```

# Reading the header

For UVI our most important reading to inspect are currently the following:

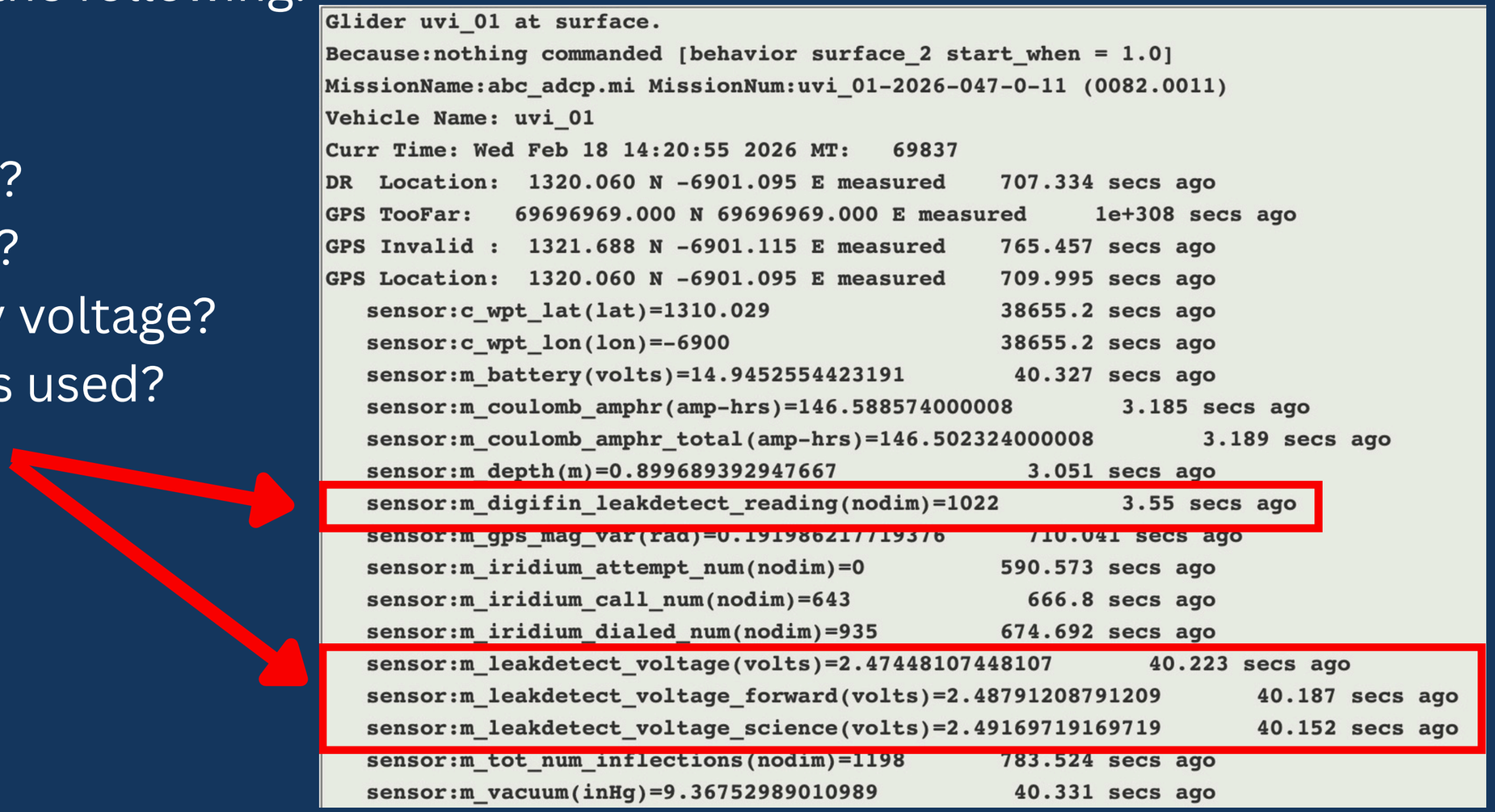
- Why did it surface
- Which glider is this?
- Where is the glider?
- What is the battery voltage?
- How many amp-hrs used?

```
Glider uvi_01 at surface.
Because:nothing commanded [behavior surface_2 start_when = 1.0]
MissionName:abc_adcp.mi MissionNum:uvi_01-2026-047-0-11 (0082.0011)
Vehicle Name: uvi_01
Curr Time: Wed Feb 18 14:20:55 2026 MT:   69837
DR  Location:  1320.060 N -6901.095 E measured    707.334 secs ago
GPS TooFar:   69696969.000 N 69696969.000 E measured     1e+308 secs ago
GPS Invalid :  1321.688 N -6901.115 E measured    765.457 secs ago
GPS Location:  1320.060 N -6901.095 E measured    709.995 secs ago
   sensor:c_wpt_lat(lat)=1310.029                 38655.2 secs ago
   sensor:c_wpt_lon(lon)=-6900                    38655.2 secs ago
   sensor:m_battery(volts)=14.9452554423191        40.327 secs ago
   sensor:m_coulomb_amphr(amp-hrs)=146.588574000008        3.185 secs ago
   sensor:m_coulomb_amphr_total(amp-hrs)=146.502324000008        3.189 secs ago
   sensor:m_depth(m)=0.899689392947667              3.051 secs ago
   sensor:m_digifin_leakdetect_reading(nodim)=1022         3.55 secs ago
   sensor:m_gps_mag_var(rad)=0.191986217719376      710.041 secs ago
   sensor:m_iridium_attempt_num(nodim)=0          590.573 secs ago
   sensor:m_iridium_call_num(nodim)=643             666.8 secs ago
   sensor:m_iridium_dialed_num(nodim)=935         674.692 secs ago
   sensor:m_leakdetect_voltage(volts)=2.47448107448107       40.223 secs ago
   sensor:m_leakdetect_voltage_forward(volts)=2.48791208791209       40.187 secs ago
   sensor:m_leakdetect_voltage_science(volts)=2.49169719169719       40.152 secs ago
   sensor:m_tot_num_inflections(nodim)=1198       783.524 secs ago
   sensor:m_vacuum(inHg)=9.36752989010989          40.331 secs ago
```

# Reading the header

For UVI our most important reading to inspect are currently the following:

- Why did it surface
- Which glider is this?
- Where is the glider?
- What is the battery voltage?
- How many amp-hrs used?
- Any sign of a leak?



# Reading the header

For UVI our most important reading to inspect are currently the following:

- Why did it surface
- Which glider is this?
- Where is the glider?
- What is the battery voltage?
- How many amp-hrs used?
- Any sign of a leak?
- Ideal internal vacuum at the surface?

```
Glider uvi_01 at surface.
Because:nothing commanded [behavior surface_2 start_when = 1.0]
MissionName:abc_adcp.mi MissionNum:uvi_01-2026-047-0-11 (0082.0011)
Vehicle Name: uvi_01
Curr Time: Wed Feb 18 14:20:55 2026 MT:   69837
DR  Location:  1320.060 N -6901.095 E measured    707.334 secs ago
GPS TooFar:   69696969.000 N 69696969.000 E measured     1e+308 secs ago
GPS Invalid :  1321.688 N -6901.115 E measured    765.457 secs ago
GPS Location:  1320.060 N -6901.095 E measured    709.995 secs ago
   sensor:c_wpt_lat(lat)=1310.029                  38655.2 secs ago
   sensor:c_wpt_lon(lon)=-6900                     38655.2 secs ago
   sensor:m_battery(volts)=14.9452554423191         40.327 secs ago
   sensor:m_coulomb_amphr(amp-hrs)=146.588574000008        3.185 secs ago
   sensor:m_coulomb_amphr_total(amp-hrs)=146.502324000008        3.189 secs ago
   sensor:m_depth(m)=0.899689392947667               3.051 secs ago
   sensor:m_digifin_leakdetect_reading(nodim)=1022         3.55 secs ago
   sensor:m_gps_mag_var(rad)=0.191986217719376      710.041 secs ago
   sensor:m_iridium_attempt_num(nodim)=0           590.573 secs ago
   sensor:m_iridium_call_num(nodim)=643              666.8 secs ago
   sensor:m_iridium_dialed_num(nodim)=935          674.692 secs ago
   sensor:m_leakdetect_voltage(volts)=2.47448107448107       40.223 secs ago
   sensor:m_leakdetect_voltage_forward(volts)=2.48791208791209       40.187 secs ago
   sensor:m_leakdetect_voltage_science(volts)=2.49169719169719       40.152 secs ago
   sensor:m_tot_num_inflections(nodim)=1198        783.524 secs ago
   sensor:m_vacuum(inHg)=9.36752989010989           40.331 secs ago
```

# Oddities, Warnings, Errors

```
^W              name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
                   [I Installed] [- Not_Installed]
                     [u In_use] [- Not_In_use] [X Out_of_Service]
            name        limits     stats (#total/#mission/#segment)
                        os w/s w/m os       errors      warnings      oddities
 0         simdrvr  -
 1     test_driver  -
 2           ARGOS* I u  -1  20   5  0
 3        WATCHDOG  I u  -1  -1  -1  0
 4         DEADMAN  I u  -1  20   5  0
 5         CONSOLE* I u  -1  20   5  0
 6             GPS  I u  -1  20   5  0 [   0   0  0] [  19   1  0] [   0   0  0]
 7    attitude_rev  I u   3  20   5  0
 8  ocean_pressure  I u   3  20   5  0
 9          vacuum  I u   3  20   5  0
10         battery  I u   3  20   5  0
11        air_pump  I u   3  20   5  0
12     pitch_motor  I u   3  20   5  0 [   0   0  0] [   0   0  0] [   6   0  0]
13   science_super  I u   3  20   5  0 [   0   0  0] [   0   0  0] [  74   0  0]
14         digifin  I u   3  20   5  0 [   0   0  0] [   5   0  0] [ 692  17  3]
15       altimeter  I u   3  20   5  0
16   altimeter_232  -
17         IRIDIUM* I u  -1  20   5  0 [   0   0  0] [   1   0  0] [ 196   6  0]
18      leakdetect  I u   3  20   5  0
19        recovery  -
```

List of compounding issues on glider components:

- Oddities are common
- Warnings can sometimes be bad
- Errors are bad

Structure:

[#  #  #]

Total # of issues

# Oddities, Warnings, Errors

```
^W              name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
                   [I Installed] [- Not_Installed]
                     [u In_use] [- Not_In_use] [X Out_of_Service]
            name        limits     stats (#total/#mission/#segment)
                        os w/s w/m os       errors      warnings      oddities
 0         simdrvr  -
 1     test_driver  -
 2           ARGOS* I u  -1  20   5  0
 3        WATCHDOG  I u  -1  -1  -1  0
 4         DEADMAN  I u  -1  20   5  0
 5         CONSOLE* I u  -1  20   5  0
 6             GPS  I u  -1  20   5  0 [   0   0  0] [  19   1  0] [   0   0  0]
 7    attitude_rev  I u   3  20   5  0
 8  ocean_pressure  I u   3  20   5  0
 9          vacuum  I u   3  20   5  0
10         battery  I u   3  20   5  0
11        air_pump  I u   3  20   5  0
12     pitch_motor  I u   3  20   5  0 [   0   0  0] [   0   0  0] [   6   0  0]
13   science_super  I u   3  20   5  0 [   0   0  0] [   0   0  0] [  74   0  0]
14         digifin  I u   3  20   5  0 [   0   0  0] [   5   0  0] [ 692  17  3]
15       altimeter  I u   3  20   5  0
16   altimeter_232  -
17         IRIDIUM* I u  -1  20   5  0 [   0   0  0] [   1   0  0] [ 196   6  0]
18      leakdetect  I u   3  20   5  0
19        recovery  -
```

List of compounding issues on glider components:

- Oddities are common
- Warnings can sometimes be bad
- Errors are bad

Structure:

[#  #  #]

Total # in mission

# Oddities, Warnings, Errors

```
^W              name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
                   [I Installed] [- Not_Installed]
                     [u In_use] [- Not_In_use] [X Out_of_Service]
            name         limits     stats (#total/#mission/#segment)
                         os w/s w/m os       errors      warnings      oddities
 0         simdrvr  -
 1     test_driver  -
 2           ARGOS* I u  -1  20   5  0
 3        WATCHDOG  I u  -1  -1  -1  0
 4         DEADMAN  I u  -1  20   5  0
 5         CONSOLE* I u  -1  20   5  0
 6             GPS  I u  -1  20   5  0 [   0   0  0] [  19   1  0] [   0   0  0]
 7    attitude_rev  I u   3  20   5  0
 8  ocean_pressure  I u   3  20   5  0
 9          vacuum  I u   3  20   5  0
10         battery  I u   3  20   5  0
11        air_pump  I u   3  20   5  0
12     pitch_motor  I u   3  20   5  0 [   0   0  0] [   0   0  0] [   6   0  0]
13   science_super  I u   3  20   5  0 [   0   0  0] [   0   0  0] [  74   0  0]
14         digifin  I u   3  20   5  0 [   0   0  0] [   5   0  0] [ 692  17  3]
15       altimeter  I u   3  20   5  0
16   altimeter_232  -
17         IRIDIUM* I u  -1  20   5  0 [   0   0  0] [   1   0  0] [ 196   6  0]
18      leakdetect  I u   3  20   5  0
19        recovery  -
```

List of compounding issues on glider components:

- Oddities are common
- Warnings can sometimes be bad
- Errors are bad

Structure:

[#  #  #]

Total # in segment

# Oddities, Warnings, Errors

```
^W              name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
                   [I Installed] [- Not_Installed]
                     [u In_use] [- Not_In_use] [X Out_of_Service]
            name        limits     stats (#total/#mission/#segment)
                        os w/s w/m os       errors      warnings      oddities
 0         simdrvr  -
 1     test_driver  -
 2           ARGOS* I u  -1  20   5  0
 3        WATCHDOG  I u  -1  -1  -1  0
 4         DEADMAN  I u  -1  20   5  0
 5         CONSOLE* I u  -1  20   5  0
 6             GPS  I u  -1  20   5  0 [   0   0  0] [  19   1  0] [   0   0  0]
 7    attitude_rev  I u   3  20   5  0
 8  ocean_pressure  I u   3  20   5  0
 9          vacuum  I u   3  20   5  0
10         battery  I u   3  20   5  0
11        air_pump  I u   3  20   5  0
12     pitch_motor  I u   3  20   5  0 [   0   0  0] [   0   0  0] [   6   0  0]
13   science_super  I u   3  20   5  0 [   0   0  0] [   0   0  0] [  74   0  0]
14         digifin  I u   3  20   5  0 [   0   0  0] [   5   0  0] [ 692  17  3]
15       altimeter  I u   3  20   5  0
16   altimeter_232  -
17         IRIDIUM* I u  -1  20   5  0 [   0   0  0] [   1   0  0] [ 196   6  0]
18      leakdetect  I u   3  20   5  0
19        recovery  -
```

Oddities and warnings most common in:

- GPS
- Iridium
- Digifin

This is okay, but still watch for errors

# Using Slocum Fleet Mission Control (SFMC): Data Visualization

# Data Visualization

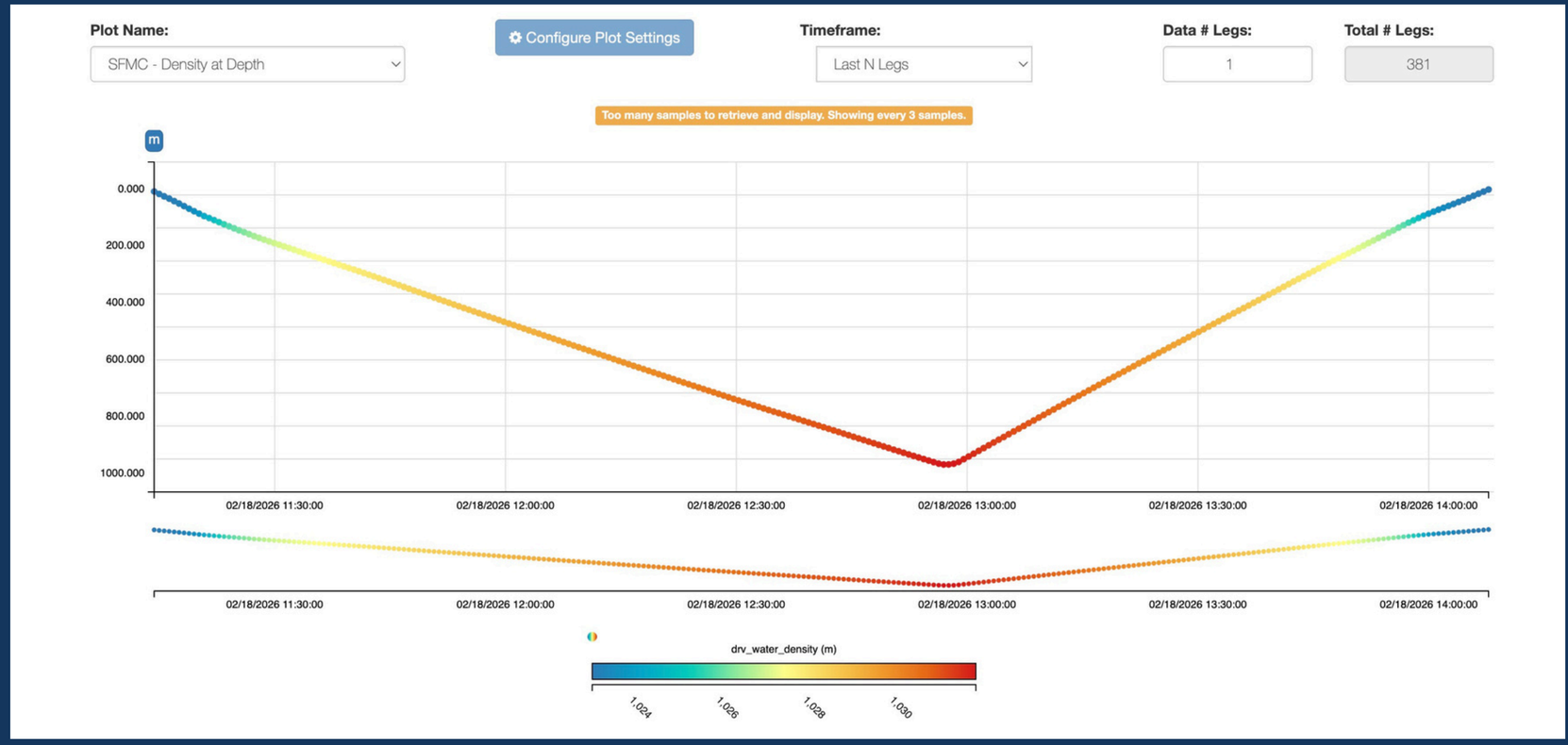
Goals:

- Is the glider flying as intended?
- Has flight performance changed recently or over time?

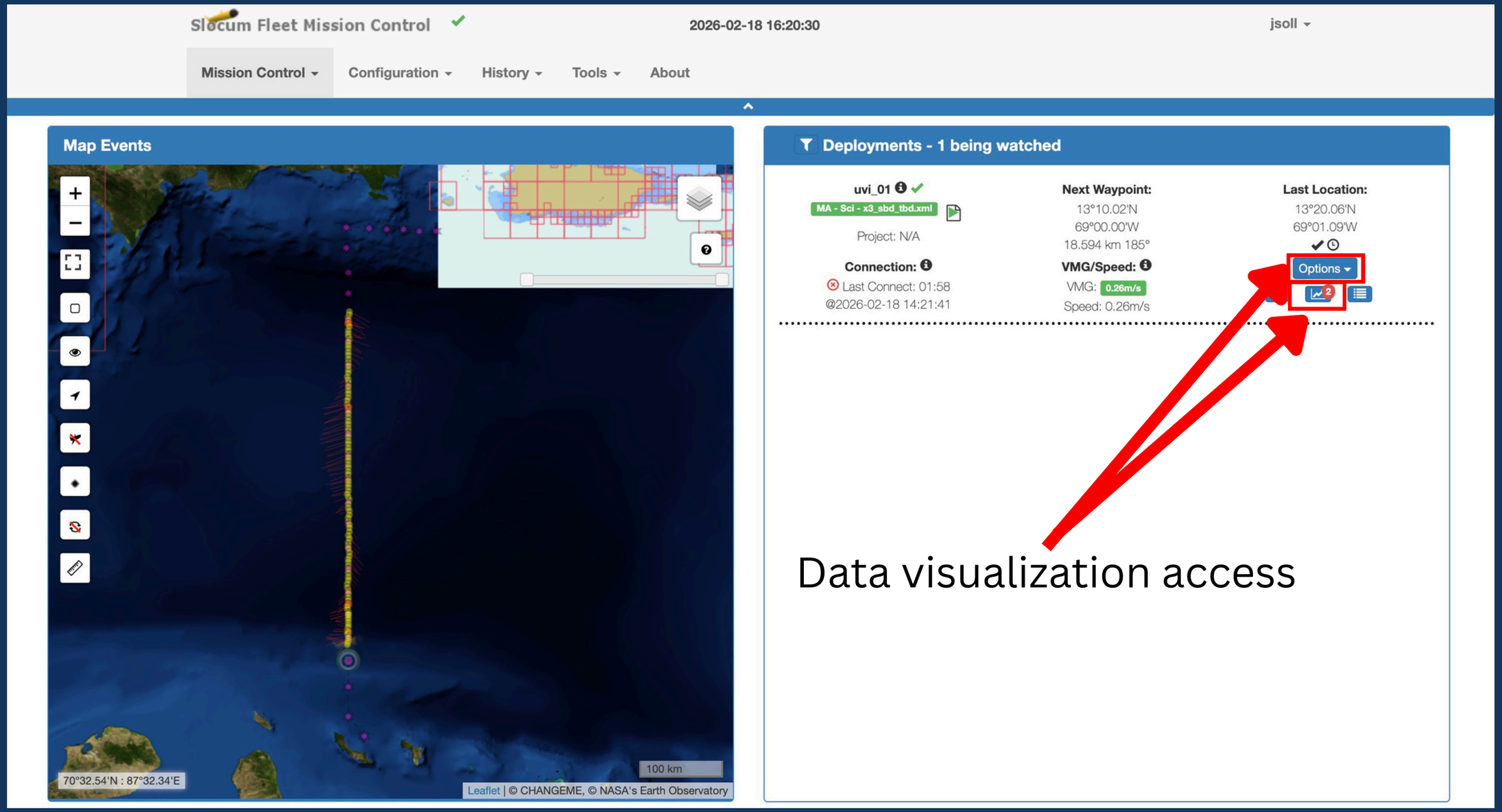
Note that the glider will only send data that it was programmed to send while in mission

Most important data:

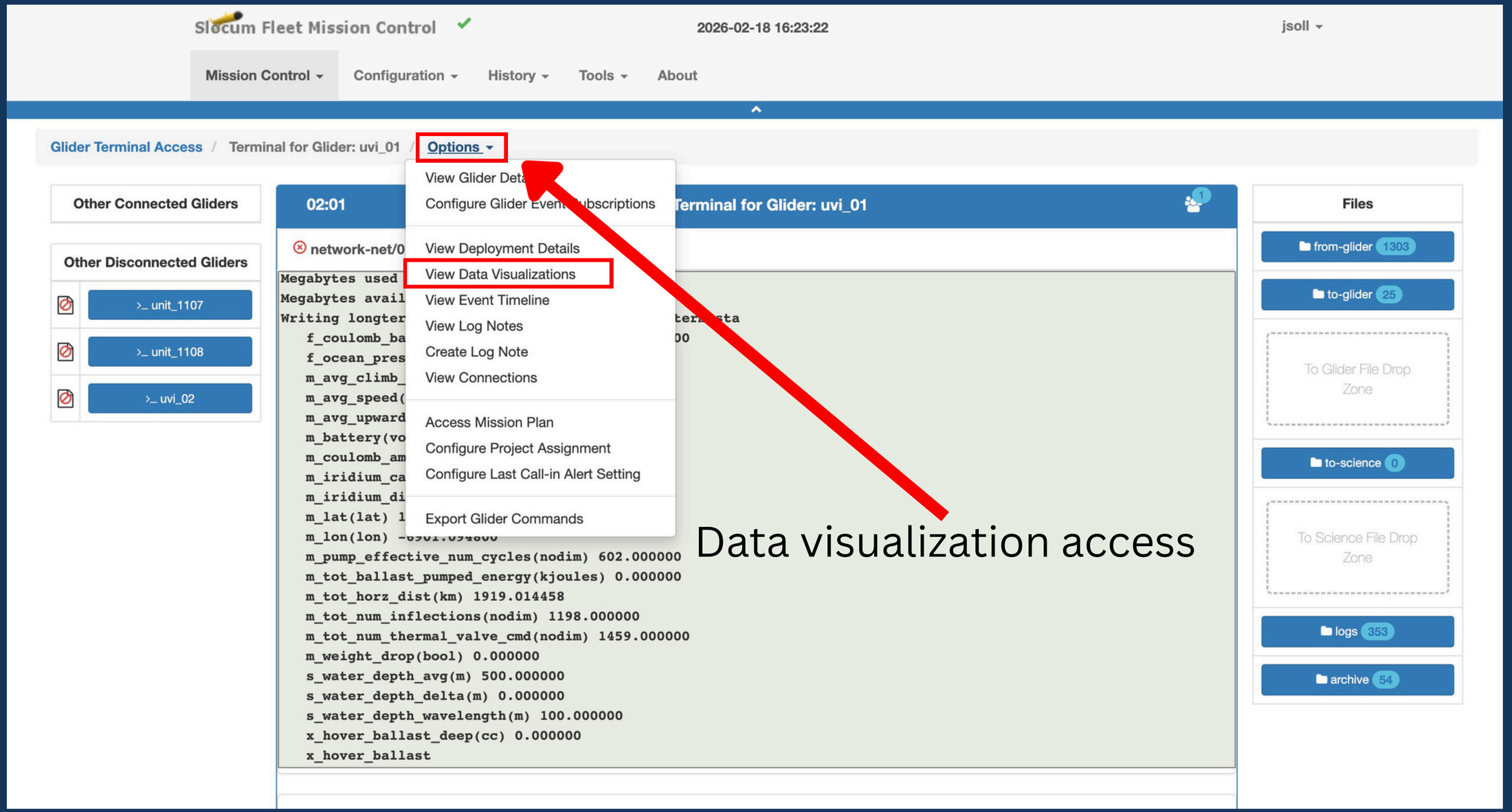
- Density at depth
- Roll
- Pitch



# Data Visualization



# Data Visualization

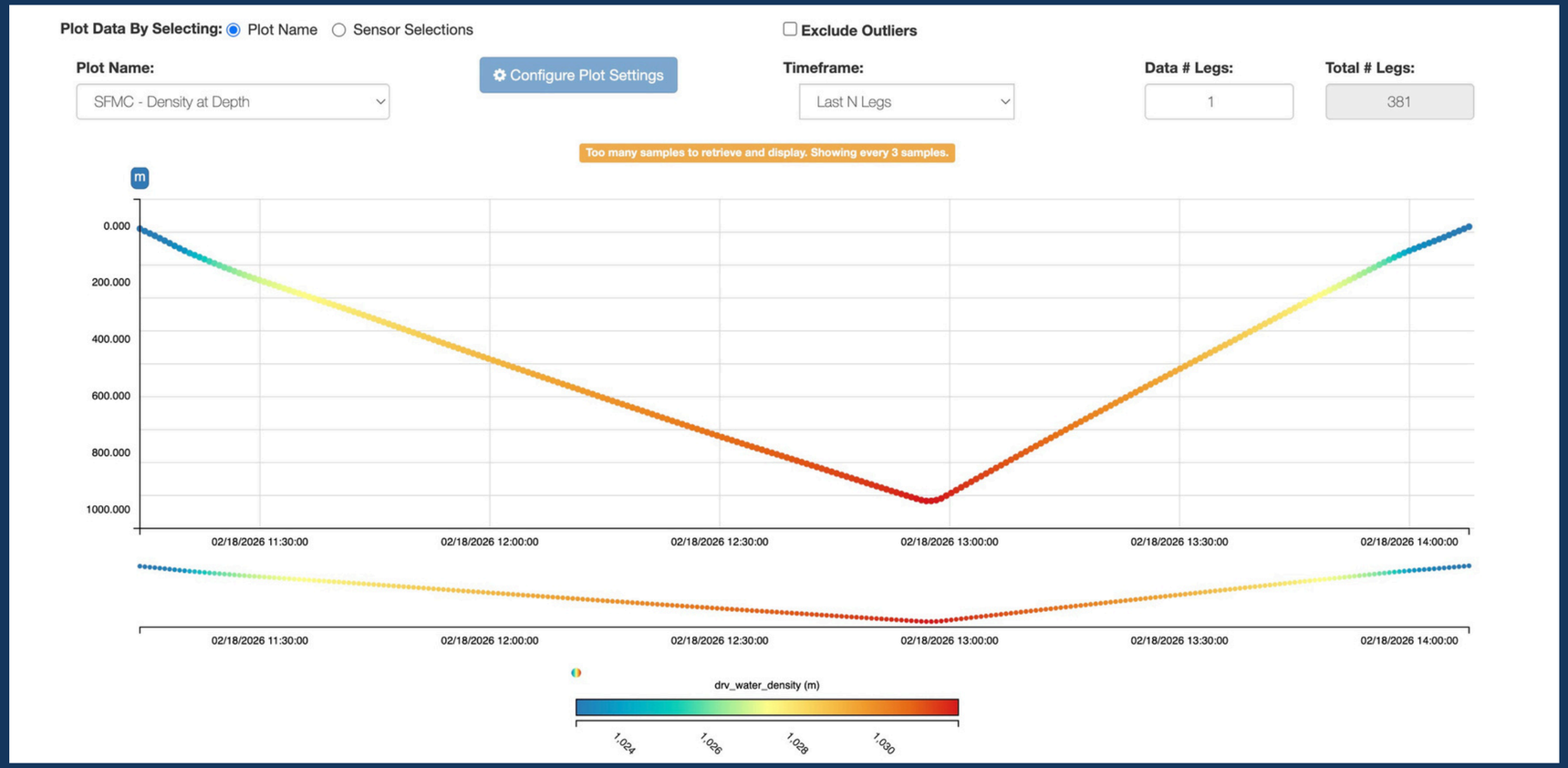


# Data Visualization

## Density at depth

Can use to check glider's depth range

Used to check density to make sure the glider's range of buoyancy can adjust for the water density to climb and dive

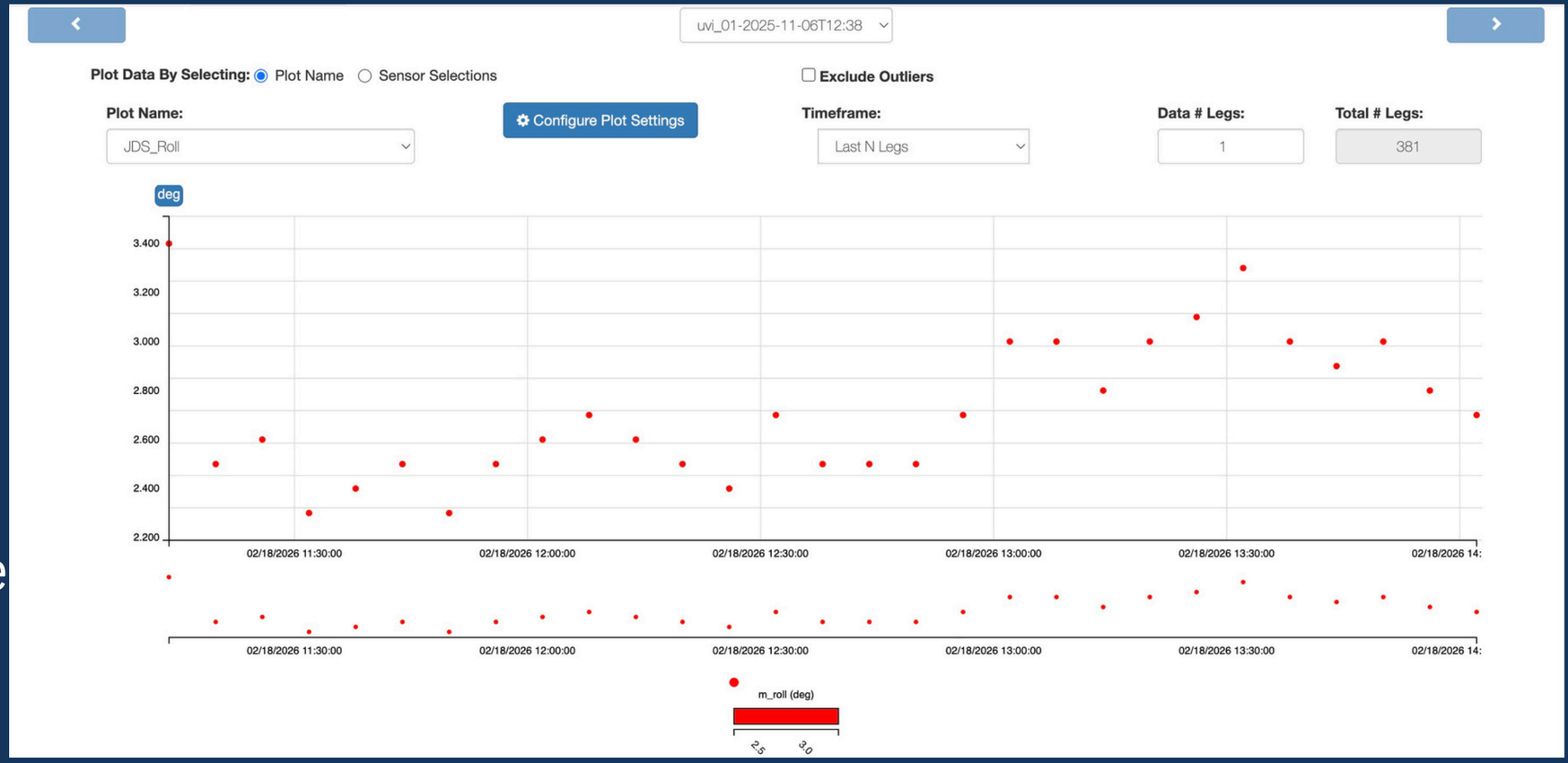


# Data Visualization

## Glider roll

Ensure that glider is within nominal range of roll (well balanced)

- (+) value is roll to starboard
- (-) value is roll to port
- Important to be within 5 degrees on either side

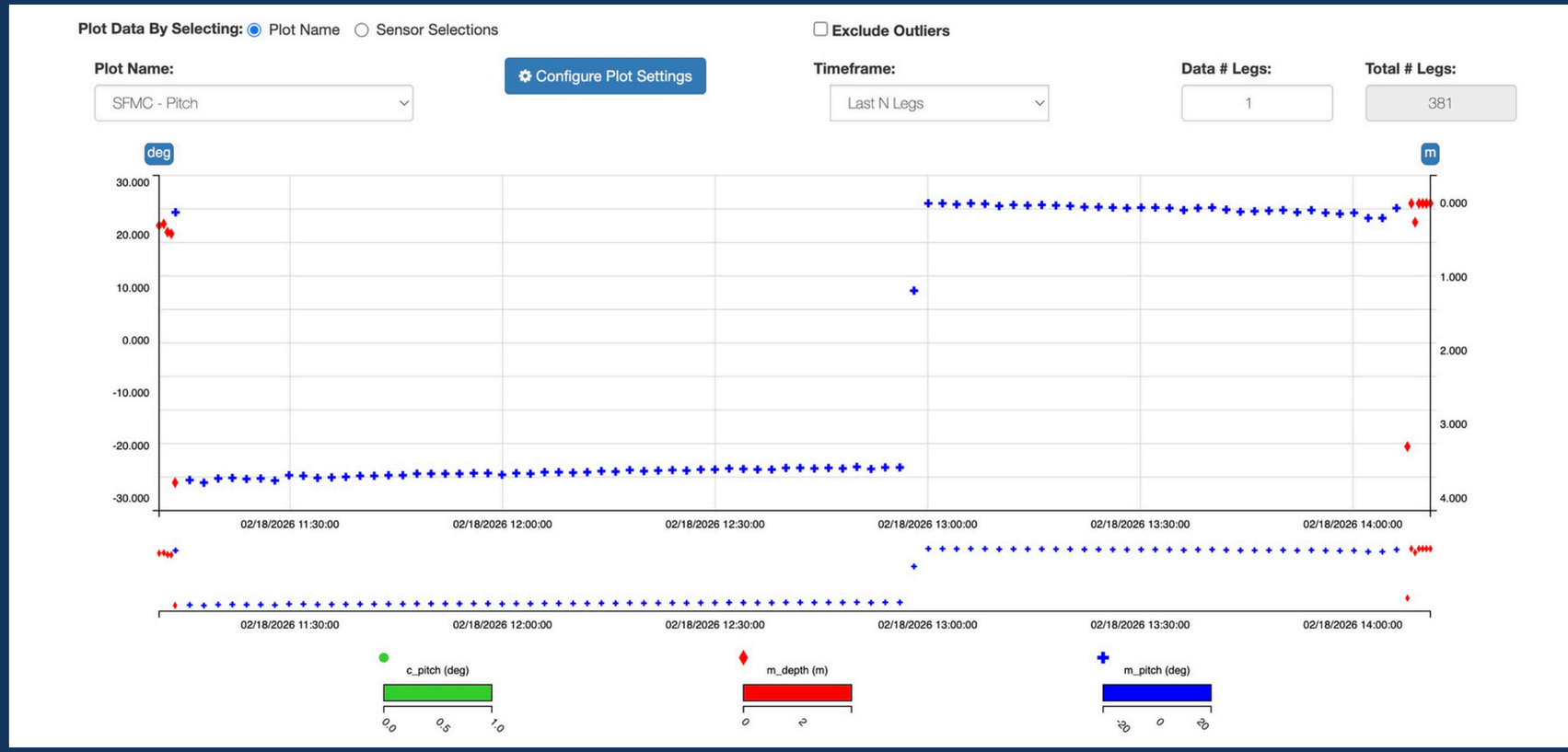


# Data Visualization

Glider pitch

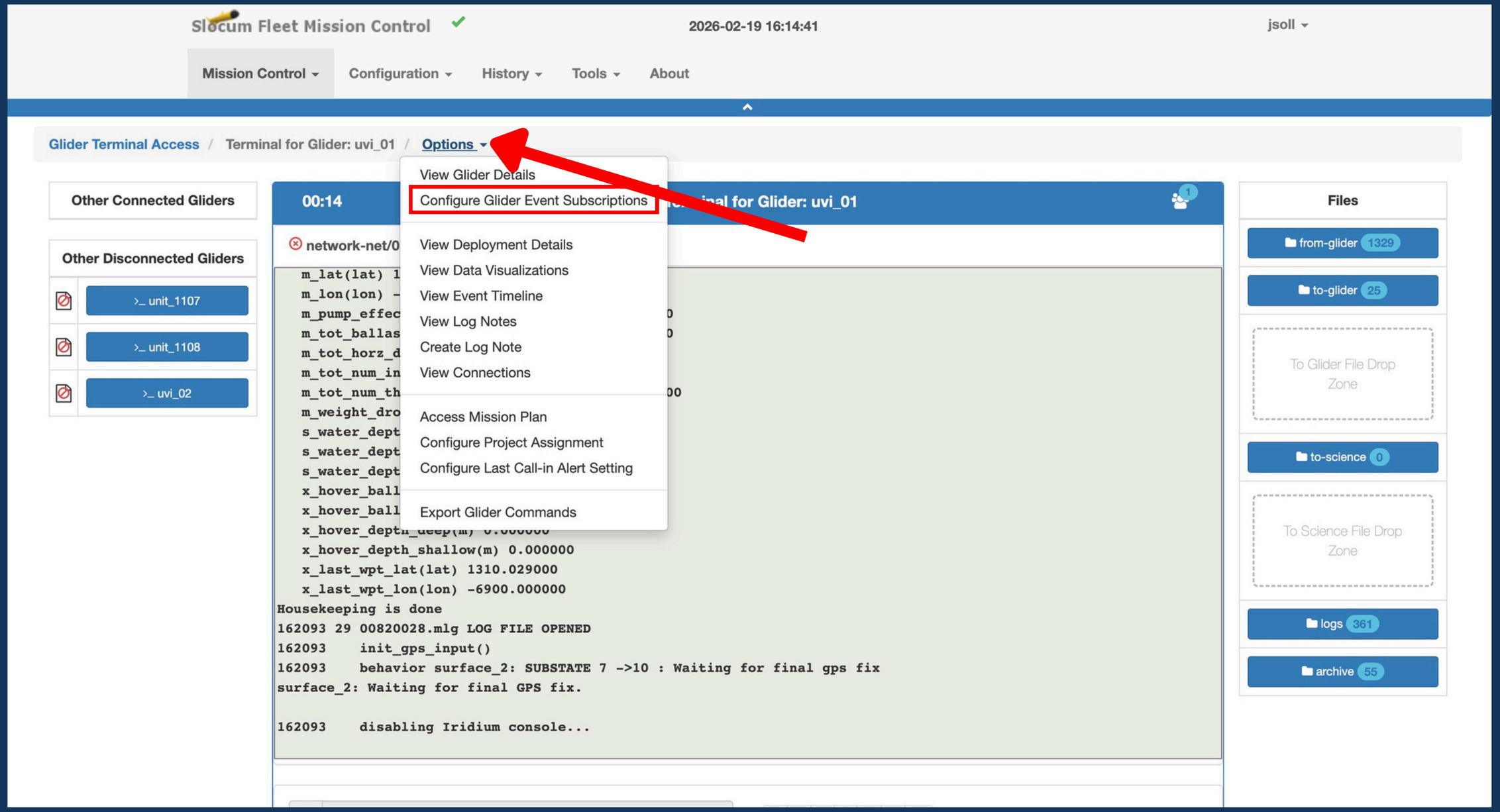
Ensure that glider is diving and climbing at reasonable angles

- Ideal angle is generally around ~26 degrees, positive (climb) or negative (dive)

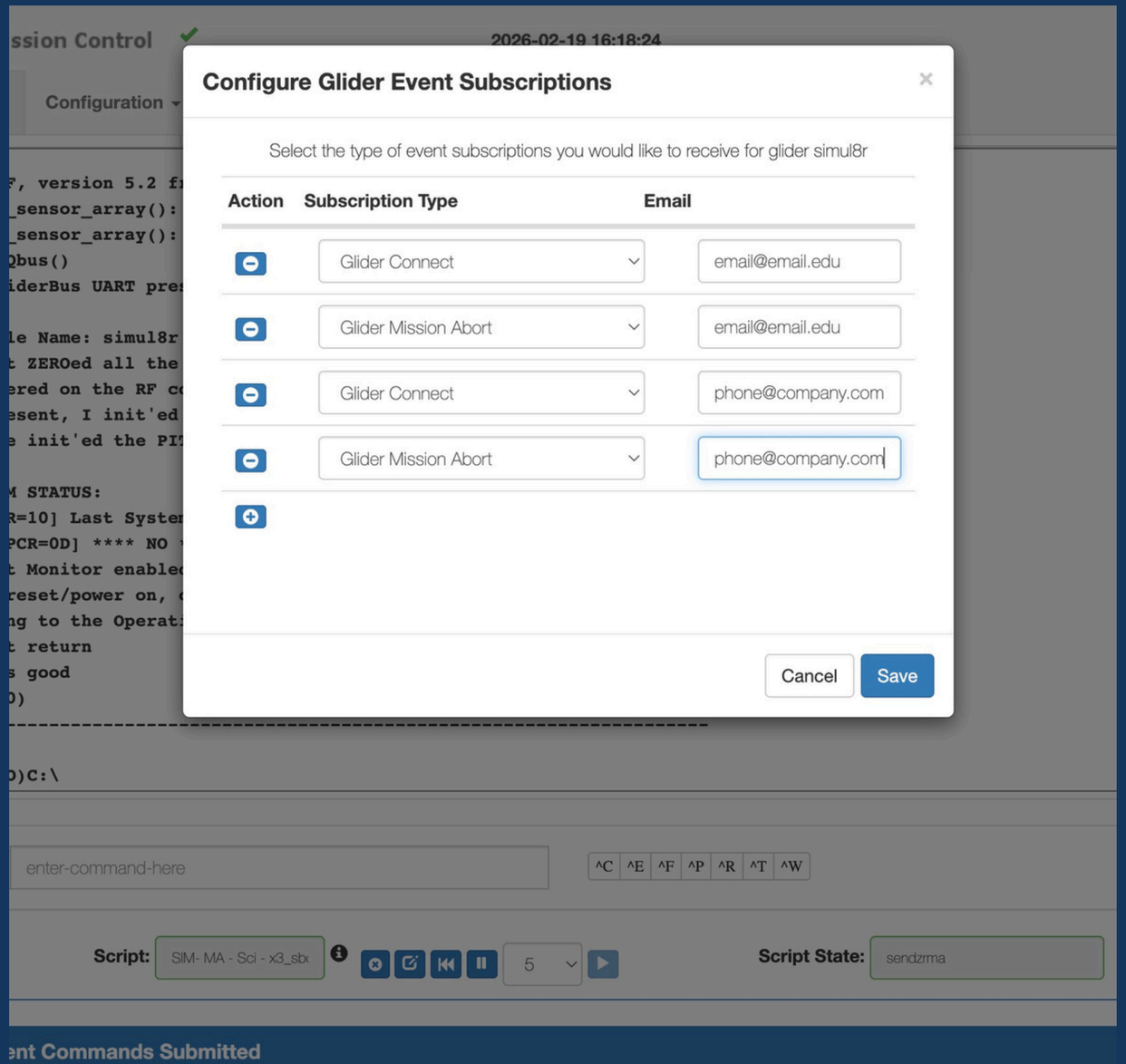


# Using Slocum Fleet Mission Control (SFMC): Setting up Notifications

# Getting Notifications



# Getting Notifications




Can set up email and phone notifications of specific glider events

Phone number notifications using carrier email to text:

- T-mobile: number@tmomail.net
- AT&T: number@txt.att.net
- Verizon: number@vtext.com
- Works for other carriers too

# Glider Aborts

# Glider Aborts

- Glider ends mission and returns to surface for some reason (not all aborts are catastrophic)

# Glider Aborts

- Glider ends mission and returns to surface for some reason (not all aborts are catastrophic)
- Three types of aborts:
  - Synchronous aborts: device errors, user-programmed behaviors

# Glider Aborts

- Glider ends mission and returns to surface for some reason (not all aborts are catastrophic)
- Three types of aborts:
  - Synchronous aborts: device errors, user-programmed behaviors
  - Out of band aborts: software cannot execute something, or a synchronous abort attempt failed

# Glider Aborts

- Glider ends mission and returns to surface for some reason (not all aborts are catastrophic)
- Three types of aborts:
  - Synchronous aborts: device errors, user-programmed behaviors
  - Out of band aborts: software cannot execute something, or a synchronous abort attempt failed
  - Hardware aborts: something in circuit not working properly

# Glider Aborts

- Glider ends mission and returns to surface for some reason (not all aborts are catastrophic)
- Three types of aborts:
  - Synchronous aborts: device errors, user-programmed behaviors
  - Out of band aborts: software cannot execute something, or a synchronous abort attempt failed
  - Hardware aborts: something in circuit not working properly
- Some aborts can be addressed and the mission resumed (i.e., glider interrupted by wave while sending data; sometimes resumes automatically)

# Glider Aborts

- Glider ends mission and returns to surface for some reason (not all aborts are catastrophic)
- Three types of aborts:
  - Synchronous aborts: device errors, user-programmed behaviors
  - Out of band aborts: software cannot execute something, or a synchronous abort attempt failed
  - Hardware aborts: something in circuit not working properly
- Some aborts can be addressed and the mission resumed (i.e., glider interrupted by wave while sending data; sometimes resumes automatically)
- Other aborts cause the glider to eject a 500g weight to ensure it surfaces, causing the end of the mission

# Glider Aborts

- When the glider aborts and returns to the surface, you will receive a message through terminal (and email/text if set up) like this:

**Glider: uvi_01 Event: Glider Mission Abort Reason: MS_ABORT_BEH_ERROR**
**ABORT HISTORY: total since reset: 13**

**ABORT HISTORY: last abort cause: MS_ABORT_BEH_ERROR**

**ABORT HISTORY: last abort details: layered_control(): Behavior surface_2 entered B_ERROR state**

**ABORT HISTORY: last abort time: 2026-02-17T18:36:12**

**ABORT HISTORY: last abort segment: uvi_01-2026-046-0-21 (0081.0021)**

**ABORT HISTORY: last abort mission: abc_adcp.mi**

# Glider Aborts

The usual process for handling aborts:

- Don’t panic

# Glider Aborts

The usual process for handling aborts:

- Don’t panic
- Find out why the glider aborted and get a new GPS fix

# Glider Aborts

The usual process for handling aborts:

- Don’t panic
- Find out why the glider aborted and get a new GPS fix
- Send commands to gain control of the glider

# Glider Aborts

The usual process for handling aborts:

- Don’t panic
- Find out why the glider aborted and get a new GPS fix
- Send commands to gain control of the glider
- Send data files to inspect possible causes

# Glider Aborts

The usual process for handling aborts:

- Don’t panic
- Find out why the glider aborted and get a new GPS fix
- Send commands to gain control of the glider
- Send data files to inspect possible causes
- Determine if the mission can be restarted or if it is a mission ending abort

# Glider Aborts

The usual process for handling aborts:

- Don’t panic
- Find out why the glider aborted and get a new GPS fix
- Send commands to gain control of the glider
- Send data files to inspect possible causes
- Determine if the mission can be restarted or if it is a mission ending abort
- Determine what needs to be done/addressed before resuming the mission after an abort (if it is safe)

# Questions?

Thank you!

joshua.soll@students.uvi.edu
travis.hamlin@uvi.edu